Polynomial-pass semi-streaming lower bound for exact degeneracy and k-cores

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Editorial note. Native fixed arXiv v1 author TeX under CC BY 4.0. Counted result thm:degen-lb includes the immediate k-core and degeneracy corollaries as one material. Full source-local MultiHPC/HPC support and every deferred appendix proof are retained. All six original author diagrams are required assets. The final paragraph on unrelated LFMIS and min-cut reductions is omitted because authors skip its formal details. Appendix input directives are expanded by selecting original-file spans; no mathematical text is supplied by the collector.

Allowed standard prerequisites (editorial).

- Set-intersection internal information lower bound. Jayram, Kumar and Sivakumar 2003; author Proposition prop:SI-hard and exact distribution retained
- Standard entropy, mutual information, rectangle property, Chernoff and Pinsker facts. Author information-theory and sampling appendix retained; Cover and Thomas 2006 and cited sources

2.1 k-Cores and Degeneracy in Graph Streams

For any undirected graph $G = (V, E)$ and integer $k \geqslant 1$, a *k-core* in G is a maximal set S of vertices such that the induced subgraph of G on S, denoted by $G[S]$, has a minimum degree of at least k. In other words, any vertex in S has at least k other neighbors in S.

k-Cores provide a natural notion of well-connectedness in massive graphs, and as such, computing k-cores (and more generally k-core decompositions; see, e.g., [LSY+22]) has been widely studied in databases [BGKV14, CZL+20, LZZ+19], social networks [DBS17, DBS18, KBVT15], machine learning [ADBV05, ELM18, GLM19], among others [LWSX18, GBGL20, SGJ+13].

As a result, in recent years, there has been a rapidly growing body of work on computing k-cores on massive graphs in parallel and streaming models of computation [SGJ+13, DBS17, DBS18, ELM18, GLM19, LSY+22]. In particular, [ELM18] presented a single-pass algorithm that for any $\varepsilon > 0$, computes a $(1-\varepsilon)$-approximation of every k-core in G (i.e., obtains a $(1-\varepsilon)$-approximate k-core decomposition) in $\widetilde{O}(n/\varepsilon^2)$ space (see also [SGJ+13] for an earlier streaming algorithm and [GLM19] for a closely related parallel algorithm).

The *degeneracy* of a graph $G = (V, E)$, denoted by $\kappa(G)$, is the largest integer $k \geqslant 0$ such that G contains a non-empty k-core. The simple greedy algorithm that at every step peels off the smallest degree vertex results in the so-called *degeneracy ordering* of G and $\kappa(G)$ is equal to the largest degree of a vertex removed in this peeling process [MB83]. Degeneracy is a standard measure of uniform sparsity and is closely related to other such notions like arboricity (which is within a factor 2 of degeneracy). Moreover, computing degeneracy is a subroutine for approximating various other problems such as arboricity [AYZ97], densest subgraph [Cha00], $(\kappa + 1)$ coloring [EH66].

The degeneracy problem, and closely related uniform-sparsity measures such as densest subgraph, have also been studied extensively in the graph streaming literature [BKV12,FT14,BHNT15, MTVV15, FT16, BCG20, AA20]. In particular, [FT14] provided an $O(\log n)$-pass semi-streaming algorithm that outputs a constant factor approximation to degeneracy, and [FT16] subsequently improved this to a single-pass $(1-\varepsilon)$-approximation in $\widetilde{O}(n/\varepsilon^2)$ space (see also [MTVV15] for densest subgraph and [BCG20, AA20] for degeneracy coloring).

In terms of lower bounds, [BCG20] prove that any *single-pass* streaming algorithm that computes the exact value of degeneracy or approximates it within an additive factor of λ requires $\Omega(n^2)$ space or $\Omega(n^2/\lambda^2)$ space respectively. Our *polynomial-pass* lower bounds for k-cores and degeneracy, now in a very strong sense, rule out the possibility of extending any prior semi-streaming algorithms computing near-optimal solutions to these problems, to compute *exactly* optimal solutions.

3 Preliminaries

3.1 The Set-Intersection Problem

Definition 3.1. The m**-bit Set-Intersection** problem is a communication problem for two players Alice and Bob. Alice is given a subset $X \subseteq [m]$ and Bob is given a subset $Y \subseteq [m]$, with the promise that $|X \cap Y| = 1$. Their goal is to learn the point of intersection $\{t\} = X \cap Y$.

Let us define a distribution $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ over inputs for Set-Intersection. This is the positive case of Razborov's original hard distribution for disjointness [Raz92].

Distribution $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ [AR20] on sets (X, Y) from the universe $\mathcal{X} = [m]$:

- Uniformly sample two disjoint sets $X', Y' \subseteq [m]$ of size $\frac{m}{4} - 1$.
- Sample $e^* \in [m] \setminus (X' \cup Y')$ uniformly at random and define $X = X' \cup \{e^*\}$ and $Y = Y' \cup \{e^*\}$.

Proposition 3.2. *The internal information complexity of* finding *the intersecting element in a* SetInt_m *instance sampled from* $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ *is* $\Omega(m)$.

Proposition 3.2 is not directly implied by standard set-intersection lower bounds, since they use distributions where X and Y never intersect. To obtain this proposition, one can (for example) apply the result of [JKS03] on the information cost of set disjointness protocols on intersecting distributions (see [ACK19] for more details).

3.2 The (Boolean) Hidden Pointer Chasing Problem

The Hidden Pointer Chasing (HPC) problem is a composition of Pointer Chasing and Set-Intersection, defined as follows.

Definition 3.3. The **Hidden Pointer Chasing** (HPC) problem is a communication problem with four players P_A, P_B, P_C, and P_D. Let $\mathcal{X} := \{x_1, ..., x_m\}$ and $\mathcal{Y} := \{y_1, ..., y_m\}$ be two disjoint universes.

- For any $x \in \mathcal{X}$, P_A and P_B are given an instance (A_x, B_x) of SetInt over the universe $\mathcal{Y}$ where $A_x \cap B_x = \{t_x\}$ for a single target element $t_x \in \mathcal{Y}$. We define $A := \{A_{x_1}, \ldots, A_{x_m}\}$ and $B := \{B_{x_1}, \ldots, B_{x_m}\}$ as the whole input to P_A and P_B, respectively.

- For any $y \in \mathcal{Y}$, P_C and P_D are given an instance (C_y, D_y) of SetInt over the universe $\mathcal{X}$ where $C_y \cap D_y = \{t_y\}$ for a single target element $t_y \in \mathcal{X}$. We define $C := \{C_{y_1}, \ldots, C_{y_m}\}$ and $D := \{D_{y_1}, \ldots, D_{y_m}\}$ as the whole input to P_C and P_D, respectively.
- We define two mappings $f_{AB} : \mathcal{X} \to \mathcal{Y}$ and $f_{CD} : \mathcal{Y} \to \mathcal{X}$ such that:

$$(a) \text{ for any } x \in \mathcal{X},\ f_{AB}(x) = t_x; \qquad (b) \text{ for any } y \in \mathcal{Y},\ f_{CD}(y) = t_y.$$

- Let $x_1 \in \mathcal{X}$ be an arbitrary fixed element known to all players. The pointers $z_0, z_1, z_2, \ldots$ are defined inductively as

$$z_0 := x_1, \quad z_1 := f_{AB}(z_0), \quad z_2 := f_{CD}(z_1), \quad z_3 := f_{AB}(z_2), \ \ldots.$$

For any integers $m, r \geqslant 1$, the **m-bit, r-step Hidden Pointer Chasing** problem, denoted by $\mathsf{HPC}_{m,r}$ is defined as the communication problem of finding the pointer z_r. Meaning, we wish to output the index $i \in [m]$ such that $z_r = x_i$ (when r is even) or $z_r = y_i$ (when r is odd).

We also define a Boolean variant of HPC as follows. In the **m-bit, r-step Boolean Hidden Pointer Chasing** problem, denoted by $\mathsf{BHPC}_{m,r}$, the goal is to output the bit $b(z_r) := i \mod 2$, where $z_r = x_i$ or y_i depending on if r is even or odd respectively.

3.3 The (Boolean) Multilayer Hidden Pointer Chasing Problem

The Multilayer Hidden Pointer Chasing problem (MHPC) combines r instances of HPC as its input in the following manner.

Definition 3.4. The **Multilayer Hidden Pointer Chasing** (MHPC) problem is a communication problem with four players P_A, P_B, P_C and P_D. Let m, r be integers $\geqslant 1$. The m-bit, r-layer MHPC problem, denoted by $\mathsf{MHPC}_{m,r}$ is defined as follows.

- For each $i \in [r]$, the players receive an HPC instance (A^i, B^i, C^i, D^i), each over the disjoint universes $\mathcal{X}$ and $\mathcal{Y}$ of size m. Let t^i_x denote the unique element in the intersection $A^i_x \cap B^i_x$, and t^i_y the unique element in $C^i_y \cap D^i_y$.
- For each $i \in [r]$, we have the maps $f^i_{AB} : \mathcal{X} \to \mathcal{Y}$ and $f^i_{CD} : \mathcal{Y} \to \mathcal{X}$, defined as:

$$(a) \text{ for any } x \in \mathcal{X},\ f^i_{AB}(x) = t^i_x; \qquad (b) \text{ for any } y \in \mathcal{Y},\ f^i_{CD}(y) = t^i_y.$$

- The pointers z_i are defined inductively as:

$$z_0 := x_1, \quad z_1 := f^1_{AB}(z_0), \quad z_2 := f^2_{CD}(z_1), \quad z_3 := f^3_{AB}(z_2), \ \ldots.$$

The goal of the problem is to find z_r, i.e., the index $i \in [m]$ such that $z_r = x_i$ (when r is even) or $z_r = y_i$ (when r is odd).

Analogous to HPC, we define the Boolean variant of MHPC as follows. In the **m-bit, r-layer Boolean** MHPC problem, denoted by $\mathsf{BMHPC}_{m,r}$, the goal is to output the bit $b(z_r) := i \mod 2$, where $z_r = x_i$ or y_i depending on if r is even or odd respectively.

We remark that C^j, D^j for odd j, and A^j, B^j for even j, are redundant (they do not affect the output of the function). They are kept for notational convenience and ease of presentation.

3.4 Misaligned r-Round Protocols

We are interested in bounded-round protocols for HPC, BHPC, MHPC and BMHPC. In each round, two of the players speak with each other (P_A with P_B, or P_C with P_D), but not with the other pair of players, with a message between pairs of players triggering the next round.

Definition 3.5 (Round of $\mathsf{HPC}_{m,r}$)**.** An r-round protocol is a protocol where the communication is organized in rounds $1, 2, \ldots, r$. A given pair of players — P_A and P_B or P_C and P_D — communicate with each other in the odd rounds (the first, third, etc rounds) and the other pair of players communicate with each other in the even rounds. Each round ends with the speaking pair of players sending a message to the other pair of players, and this marks the beginning of the next round.

It is easy to see that one can solve any of the above HPC variants by r-round protocols communicating $O(rm)$ bits, if P_A and P_B speak in the odd rounds, and P_C and P_D speak in the even rounds. We will show lower-bounds for *misaligned* r-round protocols where the *wrong* pair of players speak:

Definition 3.6. We call an r-round protocol *misaligned* if the pair of players P_C and P_D speak in the odd rounds, and P_A and P_B speak in the even rounds.

The above definitions naturally suggest that, in an r-round protocol, we should think of z_j as the pointer that the players wish to know by the end of round j. In a misaligned protocol, however, the players speaking in round j do not hold the Set-Intersection instance that needs to be solved to find z_j.

3.5 Notation and Terminology

Notation for Protocols. We use the notation $\pi_{\mathsf{Set\text{-}Int}}$, π_{BHPC}, and π_{BMHPC}, to denote protocols for Set-Intersection, BHPC and BMHPC, respectively.

In general, we will use the upright A to denote a random variable associated with A.

Now consider a situation where we choose inputs (A, B, C, D) from some distribution on inputs to $\mathsf{HPC}_{m,r}$. We then let $\mathsf{Z}_1, \ldots, \mathsf{Z}_r$ denote random variables with values $z_1, \ldots, z_r$ of Definition 3.3. We use the variable $j \in [r]$ to index the rounds of a r-round protocol. For any $j \in [r]$, we define Π_j as the set of all messages communicated in round j, and more generally Π denotes the entire protocol transcript.

For any $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, we define random variables $\mathsf{T}_x \in \mathcal{Y}$ and $\mathsf{T}_y \in \mathcal{X}$ giving the intersecting elements of the Set-Intersection instance $(\mathsf{A}_x, \mathsf{B}_x)$ and $(\mathsf{C}_y, \mathsf{D}_y)$, respectively. We define $\mathsf{E}_j = (\Pi_1, \ldots, \Pi_{j-1}, \mathsf{Z}_1, \ldots, \mathsf{Z}_{j-1})$ to include all messages exchanged before round j, together with the entire path followed by the pointers before round j. Note that during round j, the players wish to compute the intersection $A_{z_j} \cap B_{z_j}$ (or $C_{z_j} \cap D_{z_j}$), so knowing E_j should not help them to do that. (Note that E_1 is empty.)

On an input (A, B, C, D) to $\mathsf{BMHPC}_{m,r}$, we define the notation Z_i, j, Π_j, T_x^j, T_y^j analogously.

If we have random variables A, B, we will broadly use the notation $\mu(\mathsf{A} \mid \mathsf{B})$ to mean "the distribution of A, conditioned on the value of B that occurred". So, for example, if B is constant we may write $\mu(\mathsf{A})$, and this is some fixed distribution. But if B is non-constant, the distribution $\mu(\mathsf{A} \mid \mathsf{B})$ may vary depending on B, and so $\mu(\mathsf{A} \mid \mathsf{B})$ is itself also a random variable, which depends on B and can be determined by knowing B alone (even without knowing A).

Other notation. An input graph to a streaming algorithm will usually be called $G = (V, E)$ and n will denote $|V|$. The degeneracy of graph G will be denoted by $\kappa(G)$, or simply κ if the graph is clear from the context. The notation "$\log x$" stands for $\log_2 x$. For a positive integer a, the notation $[a]$ denotes the set $\{1, 2, \ldots, a\}$ and for integers $a < b$, $[a, b]$ denotes the set $\{a, a+1, \ldots, b\}$.

3.6 Positive Triangular Discrimination

Definition 3.7 (Positive triangular discrimination)**.** Given two distributions μ and ν on a sample space Ω, the *positive triangular discrimination* $\Lambda(\mu, \nu)$ is:

$$\Lambda(\mu, \nu) = \sum_{\substack{x \in \Omega \\ \mu(x) > \nu(x)}} \frac{(\mu(x) - \nu(x))^2}{\mu(x) + \nu(x)}.$$

In this paper, we work exclusively with *positive* triangular discrimination. Below we state some of the useful properties of positive triangular discrimination. We defer the proofs of the following propositions to Section B.

Proposition 3.8. $0 \leqslant \frac{\|\mu-\nu\|_1^2}{8} \leqslant \Lambda(\mu, \nu) \leqslant \frac{\|\mu-\nu\|_1}{2} \leqslant 1$.

The following property is already implicit in [Yeh20], but making it explicit now simplifies the presentation later. Note that an analogous equation for total variation distance, i.e., $\mathbb{E}_\mu[f(x)] \leqslant \|\mu - \nu\|_1 \cdot \max_{x \in x} f(x) + \cdot \mathbb{E}_\nu[f(x)]$ can be obtained almost from the definition. The following statement, however, needs a bit more work.

Proposition 3.9. *For any function* $f : \Omega \to \mathbb{R}_{\geqslant 0}$,

$$\mathop{\mathbb{E}}_{\mu}[f(x)] \leqslant \Lambda(\mu, \nu) \cdot \max_{x \in \Omega} f(x) + 6 \cdot \mathop{\mathbb{E}}_{\nu}[f(x)].$$

We will also make use of the convexity of Λ.

Proposition 3.10. *Positive triangular discrimination* $\Lambda(\mu, \nu)$ *is convex as a joint function of the two distributions* μ *and* ν.

3.7 Degeneracy and k-Cores

The definition of a k-core and the degeneracy $\kappa(G)$ of a graph G can be found in Section 2.1. Here, we define some related terms that come up often in the paper. We also state some basic facts about them that we later use in our proofs.

Given an ordering σ of vertices, we say $v \prec_\sigma u$ if u appears (somewhere) after v in σ.

Definition 3.11 (Outdegree with respect to an ordering)**.** For a graph $G = (V, E)$. given an ordering σ of vertices in V, the outdegree of a vertex v with respect to σ, denoted by $\mathrm{odeg}_\sigma(v)$, is the number of neighbors of v that appear after it in σ. Formally,

$$\mathrm{odeg}_\sigma(v) = |\{u \in N(v) : v \prec_\sigma u\}|$$

We can also think of it as orienting all edges (x, y) from x to y if $x \prec_\sigma y$. Then $\mathrm{odeg}_\sigma(v)$ is the outdegree of v in the resultant oriented graph.

Definition 3.12 (k-ordering)**.** Given a graph $G = (V, E)$, an ordering σ of vertices in V is called a k-ordering if $\forall v \in V : \mathrm{odeg}_\sigma(v) \leqslant k$.

The following fact connects vertex orderings to degeneracy.

Fact 3.13 ([MB83])**.** *The minimum number* k *for which a graph* G *admits a* k*-ordering is* $\kappa(G)$.

The above fact implies that to prove that a graph has degeneracy at most k, it suffices to exhibit a k-ordering of the nodes. We now formally define *degeneracy ordering* of a graph.

Definition 3.14 (Degeneracy ordering)**.** Given a graph $G = (V, E)$, a $\kappa(G)$-ordering of V is called a degeneracy ordering of G.

A degeneracy ordering of a graph can be obtained by running the following "peeling algorithm" on it: starting with an empty sequence σ, recursively remove the minimum degree vertex from G and append it to σ until the graph becomes empty. Again, the maximum degree that a node has when it is peeled by this algorithm is exactly $\kappa(G)$. If not, then the peeling algorithm gives a k-ordering for some $k < \kappa(G)$ which violates Fact 3.13. Hence, we get the following fact.

Fact 3.15. *Suppose the peeling algorithm on a graph G removes its nodes in the order $v_1, \ldots, v_n$. Define $G_0 := G$ and $G_i := G \setminus \{v_1, \ldots, v_i\}$ for $i \in [n]$. Then,*

$$\kappa(G) = \max_{i \in [n]} \deg_{G_{i-1}}(v_i)$$

.

4 New Lower Bounds for HPC and Multilayer HPC

The two main results of this section are an $\Omega(m^2)$ lower-bound against misaligned r-round protocols for $\mathsf{BMHPC}_{m,r}$, and an $\Omega(\frac{m^2}{r})$ lower-bound against misaligned r-round protocols for $\mathsf{BHPC}_{m,r}$. Notice that in a misaligned r-round round protocol for $\mathsf{BMHPC}_{m,r}$ or $\mathsf{BHPC}_{m,r}$, it is players P_C and P_D who begin the protocol by talking with each other. This means that the "wrong" pair of players begin to speak, in the sense that they wish to compute the value $\{z_1\} = A_{z_0} \cap B_{z_0}$, but this instance is with Alice and Bob, so they have no way to do this. So the first round of communication says nothing about z_1: the best P_C and P_D can do is send some information about all of their Set-Intersection instances, without knowing which is important. So each bit that P_C and P_D communicate with P_A and P_B in the first round, can only reveal $\frac{1}{m}$ bits of information about the average instance. So, in the next round, P_A and P_B cannot have learned what C_{z_1} or D_{z_1} are, either. But then, how can they learn z_2? The difficult situation is now reversed! This "always one step behind" situation is similar to what happens for pointer chasing [NW93, PRV99, Yeh20], except now the pointers are "hidden" behind set intersection instances.

How much communication must the players use in order to escape this unfortunate situation? In the case of $\mathsf{BMHPC}_{m,r}$, it is not clear how to do this in less than $O(m^2)$ bits of communication. One can think of two strategies: either reveal an entire layer, causing the protocol to become aligned at that layer, or solve roughly $\frac{m}{r}$ set intersections instances in each layer, which will cause the protocol to become aligned (follow the pointer of a solved instance) at any given layer with probability $\frac{1}{r}$. We will prove that these two strategies are essentially optimal.

However, in the case of $\mathsf{BHPC}_{m,r}$, it can be seen that $O(\frac{m^2}{r})$ bits of communication suffice. Indeed, P_C and P_D (who speak first) may choose $N = O(\frac{m}{r})$ of their Set-Intersection instances uniformly at random, solve them all by communicating $O(\frac{m^2}{r})$ bits, and then send the solution of all these instances to P_A and P_B. Then, in the next round, P_A and P_B begin the standard protocol for solving $\mathsf{HPC}_{m,r}$, meaning, they compute z_1 and send it back, then P_C and P_D compute z_2 and send it back, etc. If at any point P_A and P_B compute some z_j which is among the N revealed Set-Intersection instances, then they can skip a step, and learn z_{j+2} immediately without communicating. The protocol is now aligned, and can be finished within the remaining number of rounds. The probability that the path taken crosses one of the instances revealed by P_C and P_D is $\Omega(r/N)$, which can be made arbitrarily close to 1 with our choice of N.

On the other hand, one should not expect to be able to prove any significant lower-bound when r is a sufficiently large fraction of $\sqrt{n}$. Indeed, in this case, the birthday paradox says that the path

being traversed will cycle with high probability. The players may then follow the path (wasting the first round entirely) until they find a cycle, and then they will immediately learn z_k.

The intuitive explanation and the upper-bounds given above will lead us to conjecture the following, necessarily tight lower-bounds, which are the main results of this section.

Theorem 4.1. *Any misaligned r-round protocol, solving $\mathsf{BMHPC}_{m,r}$ correctly with constant probability requires $\Omega(m^2)$ bits of communication.*

Theorem 4.2. *Any misaligned r-round protocol, where $r = o(\sqrt{m})$, solving $\mathsf{BHPC}_{m,r}$ correctly with constant probability requires $\Omega(\frac{m^2}{r})$ bits of communication.*

A weaker lower-bound of $\frac{m^2}{r^2}$ for HPC was proven in [ACK19]. Our proofs of these tight lower-bounds follow a similar strategy to the earlier proof, but with a somewhat modified protocol inspired by [Yeh20]. We begin by proving Theorem 4.2 in Section 4.3, because the proof is slightly simpler, and it is enough for our streaming lower-bounds. The proof of Theorem 4.2 appears in Section 4.5. Both proofs follow a similar outline. We first show that a short, misaligned r-round protocol for BMHPC (or BHPC) must be "almost solving" an instance of Set-Intersection on one of the rounds, while revealing little information. The notion of "almost solving" will be formally defined in Section 4.1. Then, in Section 4.4, we show that any low-information protocol for "almost-solving" Set-Intersection can be used to actually solve Set-Intersection, also with low information. Putting it all together in Section 4.6, we will find that a misaligned r-round protocol for $\mathsf{BMHPC}_{m,r}$ communicating $o(m^2)$ bits (or for $\mathsf{BHPC}_{m,r}$ communicating $o(\frac{m^2}{r})$ bits), would allow us to solve Set-Intersection with $o(m)$ information cost, which is known to be impossible by [JKS03].

4.1 "Almost Solving"

The notion of "almost solving" is given precisely below. Intuitively, a protocol ε-solves Set-Intersection if the distribution of the intersection point $\{\mathsf{e}^*\} = \mathsf{A} \cap \mathsf{B}$ is sufficiently altered by the execution of the protocol. Meaning, the distribution of e^* conditioned on knowing the protocol (and possibly one of the inputs) is sufficiently far-away from the distribution of e^* without this knowledge. In [ACK19], the chosen notion of *sufficiently far* is the total variation distance. Here we will choose positive triangular discrimination.

Definition 4.3 (ε-solving Set-Intersection, with respect to positive triangular discrimination)**.** Let $\mathcal{D}$ be a distribution on inputs (A, B) for Set-Intersection. A randomized protocol π *internally ε-solves Set-Intersection* (with respect to positive triangular discrimination) on $\mathcal{D}$ if *at least one* of the following hold true (where Π is the transcript of π, including public randomness):

$$\mathop{\mathbb{E}}_{\Pi,\mathsf{A}}[\Lambda(\mu(\mathsf{e}^* \mid \Pi, \mathsf{A}), \mu(\mathsf{e}^* \mid \mathsf{A}))] \geqslant \varepsilon \qquad \text{or} \qquad \mathop{\mathbb{E}}_{\Pi,\mathsf{B}}[\Lambda(\mu(\mathsf{e}^* \mid \Pi, \mathsf{B}), \mu(\mathsf{e}^* \mid \mathsf{B}))] \geqslant \varepsilon.$$

A protocol π *externally ε-solves Set-Intersectiom* on $\mathcal{D}$ if the following holds true:

$$\mathop{\mathbb{E}}_{\Pi}[\Lambda(\mu(\mathsf{e}^* \mid \Pi), \mu(\mathsf{e}^*))] \geqslant \varepsilon.$$

Observation 4.4. *From the joint convexity of positive triangular discrimination (Proposition 3.10), it follows that for any (possibly correlated) random variables $\mathsf{A}, \mathsf{B}, \mathsf{C}$,*

$$\begin{aligned}
\mathop{\mathbb{E}}_{\mathsf{B},\mathsf{C}}[\Lambda(\mu(\mathsf{A} \mid \mathsf{B}, \mathsf{C}), \mu(\mathsf{A} \mid \mathsf{C}))] &= \mathop{\mathbb{E}}_{\mathsf{B}}\left[\sum_c \Pr[\mathsf{C} = c]\Lambda(\mu(\mathsf{A} \mid \mathsf{B}, \mathsf{C} = c), \mu(\mathsf{A} \mid \mathsf{C} = c))\right] \\
&\geqslant \mathop{\mathbb{E}}_{\mathsf{B}}\left[\Lambda\left(\sum_c \Pr[\mathsf{C} = c]\mu(\mathsf{A} \mid \mathsf{B}, \mathsf{C} = c), \sum_c \Pr[\mathsf{C} = c]\mu(\mathsf{A} \mid \mathsf{C} = c)\right)\right] \\
&= \mathop{\mathbb{E}}_{\mathsf{B}}[\Lambda(\mu(\mathsf{A} \mid \mathsf{B}), \mu(\mathsf{A}))].
\end{aligned}$$

That is, removing a random variable from both conditions can only decrease the Λ-distance. In particular, if a protocol externally ε-solves Set-Intersection on some distribution $\mathcal{D}$, it also internally ε-solves Set-Intersection on $\mathcal{D}$.

Observation 4.5. *It may be felt that a more natural definition of internally ε-solving should include any eventual private randomness of the protocol π. However, noting that $\{\mathsf{e}^*\} = \mathsf{A} \cap \mathsf{B}$ is a function of A and B, and that A and B are independent given Π (by Fact E.9) it follows that the distribution of e^* is independent of the private randomness of π, when given A and Π. So*

$$\mathop{\mathbb{E}}_{\Pi,\mathsf{A}}[\Lambda(\mu(\mathsf{e}^* \mid \Pi, \mathsf{A}), \mu(\mathsf{e}^* \mid \mathsf{A}))] = \mathop{\mathbb{E}}_{\Pi,\mathsf{A}}[\Lambda(\mu(\mathsf{e}^* \mid \Pi, \mathsf{R}_A, \mathsf{A}), \mu(\mathsf{e}^* \mid \mathsf{A}))]$$

where R_A is Alice's randomness, and likewise on Bob's side. That is, including the private randomness gives the same definition.

4.2 Hard Distributions

Next, we define two hard distributions for $\mathcal{D}_{\mathsf{BMHPC}}$ and $\mathcal{D}_{\mathsf{BHPC}}$, over inputs to BMHPC and BHPC, respectively.

Distribution $\mathcal{D}_{\mathsf{BMHPC}}$ on tuples $(\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D})$ from the universes $\mathcal{X}$ and $\mathcal{Y}$:

1. For any $i \in [r]$, and any $x \in \mathcal{X}$, sample $(A_x^i, B_x^i) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$ from the universe $\mathcal{Y}$ *independently.*
2. For any $i \in [r]$, and any $y \in \mathcal{Y}$, sample $(C_y^i, D_y^i) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$ from the universe $\mathcal{X}$ *independently.*

Distribution $\mathcal{D}_{\mathsf{BHPC}}$ [ACK19] on tuples $(\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D})$ from the universes $\mathcal{X}$ and $\mathcal{Y}$:

1. For any $x \in \mathcal{X}$, sample $(A_x, B_x) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$ from the universe $\mathcal{Y}$ *independently.*
2. For any $y \in \mathcal{Y}$, sample $(C_y, D_y) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$ from the universe $\mathcal{X}$ *independently.*

It was shown in [ACK19] that misaligned randomized r-round protocols for $\mathsf{HPC}_{m,r}$ need $\frac{m^2}{r^2}$ bits of communication under $\mathcal{D}_{\mathsf{BHPC}}$. In Section 4.5, we improve this result and show that misaligned randomized r-round protocols for $\mathsf{BHPC}_{m,r}$ (an easier problem than $\mathsf{HPC}_{m,r}$) need $\frac{m^2}{r}$ bits of communication.

We restate the following simple observations from [ACK19] about $\mathcal{D}_{\mathsf{BHPC}}$. Analogous statements can be made for $\mathcal{D}_{\mathsf{BMHPC}}$.

Observation 4.6 ([ACK19])**.** *Distribution $\mathcal{D}_{\mathsf{BHPC}}$ is <u>not</u> a product distribution. However, in this distribution:*

(i) The inputs to P_A and P_B are independent of the inputs to P_C and P_D, i.e., $(\mathsf{A}, \mathsf{B}) \perp (\mathsf{C}, \mathsf{D})$.

(ii) For any $x \in \mathcal{X}$, $(\mathsf{A}_x, \mathsf{B}_x)$ is independent of all other $(\mathsf{A}_{x'}, \mathsf{B}_{x'})$ for $x' \neq x \in \mathcal{X}$. Similarly for all $y, y' \in \mathcal{Y}$ and $(\mathsf{C}_y, \mathsf{D}_y)$ and $(\mathsf{C}_{y'}, \mathsf{D}_{y'})$.

Based on this observation, we also have the following simple property.

Proposition 4.7 ([ACK19]). *Let π_{BHPC} be any deterministic protocol for $\mathsf{BHPC}_{m,r}$ on $\mathcal{D}_{\mathsf{BHPC}}$. Then, for any transcript Π of π_{BHPC}, $(\mathsf{A},\mathsf{B}) \perp (\mathsf{C},\mathsf{D}) \mid \Pi = \Pi$.*

Proof. Follows from the rectangle property of the protocol π_{BHPC} (Fact E.9). In particular, the same exact argument as in the two-player case implies that if $[(A,B),(C,D)]$ and $[(A',B'),(C',D')]$ are mapped to the same transcript Π, then $[(A,B),(C',D')]$ and $[(A',B'),(C,D)]$ are mapped to Π as well. Hence, since $(\mathsf{A},\mathsf{B}) \perp (\mathsf{C},\mathsf{D})$ by Observation 4.6, the inputs corresponding to the same protocol would also be independent of each other, namely, $(\mathsf{A},\mathsf{B}) \perp (\mathsf{C},\mathsf{D}) \mid \Pi = \Pi$. □

4.3 Short Protocols for BMHPC Give Low-Information ε-Solvers for Set-Int

We now show that if we have a short, r-round protocol for $\mathsf{BMHPC}_{m,r}$ on $\mathcal{D}_{\mathsf{BMHPC}}$, then during one of its rounds it must be externally (and hence also internally) ε-solving Set-Intersection on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$, while revealing little information. This is proven by making precise the following intuitive line of thinking, the statement of which is formally captured by Lemma 4.11.

- By the end of round j, the speaking pair of players would like to have learned $\{\mathsf{Z}_j\} = \mathsf{A}^j_{\mathsf{Z}_{j-1}} \cap \mathsf{B}^j_{\mathsf{Z}_{j-1}}$ (for odd j, or $\{\mathsf{Z}_j\} = \mathsf{C}^j_{\mathsf{Z}_{j-1}} \cap \mathsf{D}^j_{\mathsf{Z}_{j-1}}$, for even j). Now suppose that in this round they have very little information about Z_{j-1}. Let us say, for example, that the distribution of $\mathsf{Z}_{j-1} \mid \mathsf{E}_{j-1}, \Pi_{j-1}$ is close to uniform. This certainly holds when $j = 2$, since $\mu(\mathsf{Z}_1|\mathsf{E}_1,\Pi_1) = \mu(\mathsf{Z}_1)$ is completely uniform (because the wrong pair of players speak first and the way we construct $\mathcal{D}_{\mathsf{BMHPC}}$).
- Now suppose that despite this, somehow, the two speaking players nonetheless manage to learn a lot of information about Z_j. Then they have "almost solved" an *unknown* instance (an instance belonging to the other players, no less), out of the possible remaining instances of Set-Intersection. Because they have little information about Z_{j-1}, meaning Z_{j-1} is close to uniform, they have thus "almost solved" an unknown, nearly-uniformly-chosen instance.
- Then, intuitively, they must have "almost solved" this unknown instance of Set-Intersection without revealing a lot of information about it. This turns out to be impossible, as we will see in Section 4.4.

One difficulty in implementing this intuitive strategy is in finding the right measure of information that allows for one to carry through the entire argument. In [ACK19], *total variation distance from uniform* was used as a measure of how much the speaking players know about which Set-Intersection instance they are trying to solve. But TVD incurs the quadratic loss inherent in Pinsker's inequality, leading to a lower-bound of $\frac{m^2}{r^2}$. The same quadratic loss appeared originally in Nisan and Wigderson's lower-bounds for pointer chasing [NW93], and this led to a three-decade-long gap between known upper and lower-bounds for pointer chasing. This gap was finally closed by Amir Yehudayoff [Yeh20], using positive triangular discrimination. This suggested the possibility of using Yehudayoff's approach to improve the lower-bound of [ACK19], and it is this plan that we realize here.

The following lemma states that no low-information protocol (in the sense of information cost) for BMHPC can reveal a lot of information (in the sense of Λ-distance-from-uniform) about a uniformly chosen SetInt instance at any layer j. The proof shows that any such protocol would give us a low-information protocol for ε-solving SetInt, which we will show is impossible in Section 4.4.

Lemma 4.8. *There is a constant $c > 0$ such that, for any protocol for $\mathsf{BMHPC}_{m,r}$ running on input $(\mathsf{A},\mathsf{B},\mathsf{C},\mathsf{D})$ sampled from $\mathcal{D}_{\mathsf{BMHPC}}$, and for any $j \in [r]$, we have:*

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_\mathcal{X}} \left[\Lambda(\mu(\mathsf{T}_x^j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{T}_x^j))\right] \leqslant c \cdot \frac{\mathbb{I}(\Pi\,;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B}) + \mathbb{I}(\Pi\,;\mathsf{B}^j \mid \mathsf{B}^{>j},\mathsf{A})}{m^2}$$

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{y\sim\mathcal{U}_\mathcal{Y}} \left[\Lambda(\mu(\mathsf{T}_y^j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{T}_y^j))\right] \leqslant c \cdot \frac{\mathbb{I}(\Pi\,;\mathsf{C}^j \mid \mathsf{C}^{<j},\mathsf{D}) + \mathbb{I}(\Pi\,;\mathsf{D}^j \mid \mathsf{D}^{>j},\mathsf{C})}{m^2}$$

Proof. Take any given $j \in [r]$, and consider the following protocol for ε-solving SetInt on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$:

Protocol π_{SI}: The protocol for ε-solving Set-Int using a protocol π_{HPC} for $\mathsf{BMHPC}_{m,r}$.
Input: An instance $(X,Y) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$ with $X, Y \subseteq \mathcal{Y}$.

1. **Sampling the instance.** Alice and Bob create an instance (A,B,C,D) of $\mathsf{BMHPC}_{m,r}$ as follows:
   (a) Using public coins, Alice and Bob sample an index $I \in [m]$ uniformly at random, and Alice sets $A^j_{x_I} = X$ and Bob sets $B^j_{x_I} = Y$ using their given inputs in Set-Int.
   (b) Using public coins, Alice and Bob sample $A^j_{x_1},\ldots,A^j_{x_{I-1}}$ from the A-side marginal of $\mathcal{D}_{\mathsf{Set\text{-}Int}}$, and they sample $B^j_{x_{I+1}},\ldots,B^j_{x_m}$ from the B-side marginal of $\mathcal{D}_{\mathsf{Set\text{-}Int}}$.
   (c) Using private coins, Alice samples $A^j_{x_r}$ for $r \in \{I+1,\ldots,n\}$ so that $(A^j_{x_r}, B^j_{x_r}) \sim \mathcal{D}_{\mathsf{Set\text{-}Int}}$. Similarly Bob samples $B^j_{x_\ell}$ for $\ell \in \{1,\ldots,I-1\}$. This completes the construction of (A^j, B^j).
   (d) Using public coins, Alice and Bob sample $A^{<j}, B^{>j}, C$ and D from their respective marginals in $\mathcal{D}_{\mathsf{BMHPC}}$ (this is possible by independence).
   (e) Using private coins, Alice samples $A^{>j}$ conditioned on the publicly sampled $B^{>j}$, and Bob samples $B^{<j}$ conditioned on the publicly sampled $A^{<j}$.
2. The players now run the first j rounds of the protocol π_{BMHPC} on the instance (A,B,C,D), by Alice playing P_A, Bob playing P_B, and both Alice and Bob simulating P_C and P_D with no communication (this is possible as both Alice and Bob know (C,D) entirely).
   Let us write $\Pi_{\mathsf{Set\text{-}Int}}$ to denote the messages of P_A and P_B, which were actually sent by Alice and Bob, and $\Pi_{\leqslant j}$ to denote the entire simulated transcript of π_{BMHPC}, including the messages of P_C and P_D (which Alice and Bob learned without communicating).

From a simple direct-sum argument, we get an upper-bound on the information cost of $\pi_{\mathsf{Set\text{-}Int}}$:

Claim 4.9.

$$\mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) = \frac{\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B}) + \mathbb{I}(\Pi\,;\mathsf{B}^j \mid \mathsf{B}^{>j},\mathsf{A})}{m}.$$

Proof. The internal information cost of $\pi_{\mathsf{Set\text{-}Int}}$ is

$$\mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) := \mathbb{I}(\mathsf{X}\,;\Pi_{\mathsf{SI}} \mid \mathsf{Y},\mathsf{R},\mathsf{R}_B) + \mathbb{I}(\mathsf{Y}\,;\Pi_{\mathsf{SI}} \mid \mathsf{X},\mathsf{R},\mathsf{R}_A),$$

where R, R_A, R_B are the public, Alice's, and Bob's randomness. Let us bound the first term, as the second term can be bounded in just the same way. We then derive:

$$\begin{aligned}
\mathbb{I}(\Pi_{\mathsf{Set\text{-}Int}}\,;\mathsf{X} \mid \mathsf{Y},\mathsf{R},\mathsf{R}_B) &= \mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{X} \mid \mathsf{Y},\mathsf{I},\mathsf{A}^j_{<\mathsf{I}},\mathsf{B}^j_{>\mathsf{I}},\mathsf{A}^{<j},\mathsf{B}^{>j},\mathsf{C},\mathsf{D},\mathsf{B}^j_{<\mathsf{I}},\mathsf{B}^{<j}) \\
&= \mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j_{\mathsf{I}} \mid \mathsf{I},\mathsf{A}^j_{<\mathsf{I}},\mathsf{A}^{<j},\mathsf{B},\mathsf{C},\mathsf{D}) \\
&= \sum_{i=1}^{m} \frac{1}{m}\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j_i \mid \mathsf{I}=i,\mathsf{A}^j_{<i},\mathsf{A}^{<j},\mathsf{B},\mathsf{C},\mathsf{D}) \\
&= \sum_{i=1}^{m} \frac{1}{m}\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j_i \mid \mathsf{A}^j_{<i},\mathsf{A}^{<j},\mathsf{B},\mathsf{C},\mathsf{D}) \\
&= \frac{1}{m}\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B},\mathsf{C},\mathsf{D}) \\
&= \frac{1}{m}\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B}).
\end{aligned}$$

All equalities are by definition, except for the fourth which is because none of the remaining distributions depend on the choice of I, and the last, which is because C,D are independent of A^j, even when conditioned on $\Pi_{\leqslant j}, \mathsf{A}^{<j}, \mathsf{B}$. □

On the other hand, we also get an ε-solver, for ε being exactly the Λ distance-from-uniform of a uniformly chosen coordinate in the j-th layer:

Claim 4.10. *$\pi_{\mathsf{Set\text{-}Int}}$ internally ε-solves Set-Intersection on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ where:*

$$\varepsilon \geqslant \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_{\mathcal{X}}} \left[\Lambda(\mu(\mathsf{T}^j_x \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{T}^j_x))\right].$$

Proof. By Definition 4.3 and Observation 4.5, we have an ε-solver on Bob's side, for

$$\varepsilon = \mathbb{E}[\Lambda(\mu(\mathsf{T} \mid \mathsf{Y},\Pi_{\mathsf{Set\text{-}Int}},\mathsf{I},\mathsf{R},\mathsf{R}_B),\mu(\mathsf{T} \mid \mathsf{Y},\mathsf{I},\mathsf{R},\mathsf{R}_B))],$$

where $T = X \cap Y = A^j_I \cap B^j_I$, I and $R = A^j_{<I}, B^j_{>I}, A^{<j}, B^{>j}, C, D$ are the public randomness, and $R_B = B^j_{<I}, B^{<j}$ is Bob's private randomness. This is exactly equal to

$$\mathbb{E}[\Lambda(\mu(\mathsf{T} \mid \mathsf{Y},\mathsf{E}_j,\Pi_j,\mathsf{I},\mathsf{R},\mathsf{R}_B),\mu(\mathsf{T} \mid \mathsf{Y},\mathsf{I},\mathsf{R},\mathsf{R}_B))],$$

because E_j, Π_j is a function of $\Pi_{\mathsf{Set\text{-}Int}}, I, R, R_B$ (Bob knows both $A^{<j}$, $B^{<j}$ and C, D, so he also knows $Z_{<j}$, and $\Pi_{\leqslant j}$). In turn, this equals

$$\mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j,i\sim\mathcal{U}_{[m]}} \mathbb{E}\left[\Lambda(\mu(\mathsf{T}^j_{x_i} \mid \mathsf{E}_j,\Pi_j,\mathsf{I}=i,\mathsf{Y},\mathsf{R},\mathsf{R}_B),\mu(\mathsf{T}^j_{x_i} \mid \mathsf{Y},\mathsf{I}=i,\mathsf{R},\mathsf{R}_B))\right]$$

Observe that none of the distributions above depend on the event $\mathsf{I}=i$. So this quantity is equal to

$$\mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j,i\sim\mathcal{U}_{[m]}} \mathbb{E}\left[\Lambda(\mu(\mathsf{T}^j_{x_i} \mid \mathsf{E}_j,\Pi_j,\mathsf{Y},\mathsf{R},\mathsf{R}_B),\mu(\mathsf{T}^j_{x_i} \mid \mathsf{Y},\mathsf{R},\mathsf{R}_B))\right]$$

Now, by Observation 4.4, this is greater or equal to

$$\mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j,i\sim\mathcal{U}_{[m]}} \mathbb{E}\left[\Lambda(\mu(\mathsf{T}^j_{x_i} \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{T}^j_{x_i}))\right].$$ □

From Corollary 4.13 proven in the next section, it now follows:

$$\begin{aligned}\frac{\mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B}) + \mathbb{I}(\Pi_{\leqslant j}\,;\mathsf{B}^j \mid \mathsf{B}^{>j},\mathsf{A})}{m} &\geqslant c\cdot\varepsilon m \\ &\geqslant cm\cdot \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_{\mathcal{X}}} \left[\Lambda(\mu(\mathsf{T}^j_x \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{T}^j_x))\right].\end{aligned}$$

This concludes the proof of Lemma 4.8. □

It is now possible to prove that there is no low-information protocol for solving BMHPC. The argument was roughly outlined above: if we assume by induction that the Λ distance-from-uniform is small for the answer at the previous layer, then we are essentially trying to solve a uniformly random instance at the current layer. And we just proved that this is not possible.

Lemma 4.11. *For the same constant $c>0$ of the Lemma 4.8, and for any protocol for $\mathsf{BMHPC}_{m,r}$ running on input (A,B,C,D) sampled from $\mathcal{D}_{\mathsf{BMHPC}}$, the following statement holds for all $j\in[r]$:*[1]

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{Z}_j))] \leqslant 6c\cdot\frac{\mathbb{I}(\Pi\,;\mathsf{A}^{\leqslant j},\mathsf{C}^{\leqslant j} \mid \mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi\,;\mathsf{B}^{\leqslant j},\mathsf{D}^{\leqslant j} \mid \mathsf{B}^{>j},\mathsf{D}^{>j},\mathsf{A},\mathsf{C})}{m^2}.$$

This is proven by induction. The base case is easy to see: note that $\mu(\mathsf{Z}_1 \mid \mathsf{E}_1,\Pi_1)=\mu(\mathsf{Z}_1)$ as Z_1 is independent of $\mathsf{E}_1=z_0$ which is fixed and Z_1 is independent of Π_1 as Π_1 is a function of (C,D). We obtain case j from case $j-1$ in two steps: we first show that if the protocol reveals a lot of information about the variable Z_j, then it must essentially be ε-solving a *uniformly random instance*. This is because $Z_j = T_{Z_{j-1}}$ and Z_{j-1} is, under the inductive case $j-1$, close to uniform. We then appeal to the previous lemma.

Proof. The information—measured by Λ-distance to uniform—that the j-th round of the protocol reveals about $\mathsf{Z}_j = \mathsf{T}_{\mathsf{Z}_{j-1}}$ is the information revealed about an instance sampled according to Z_{j-1}:

$$\begin{aligned}\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\left[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{Z}_j))\right] &= \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\left[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j),\mu(\mathsf{Z}_j))\right] \\ &= \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j},\Pi_{<j})}\left[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{Z}_{<j},\Pi_{<j}),\mu(\mathsf{Z}_j))\right] \\ &\qquad\text{(by definition of } \mathsf{E}_j := (\mathsf{Z}_{<j},\Pi_{<j}), \text{ and since } \mathsf{Z}_j \text{ and } \Pi_j \text{ are independent given } \mathsf{E}_j.) \\ &= \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j},\Pi_{<j})}\left[\Lambda(\mu(\mathsf{T}^j_{\mathsf{Z}_{j-1}} \mid \mathsf{Z}_{<j-1},\mathsf{Z}_{j-1},\Pi_{<j}),\mu(\mathsf{Z}_j))\right] \\ &\qquad\qquad\text{(by definition, the pointer } Z_j = T_{Z_{j-1}}) \\ &= \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j-1},\Pi_{<j})} \mathop{\mathbb{E}}_{z\sim\mathsf{Z}_{j-1}\mid(\mathsf{Z}_{<j-1},\Pi_{<j})}\left[\Lambda(\mu(\mathsf{T}^j_z \mid \mathsf{Z}_{<j-1},\Pi_{<j}),\mu(\mathsf{Z}_j))\right].\end{aligned}$$

We removed the conditioning on $\mathsf{Z}_{j-1}=z$ in the first distribution because, conditioned on knowing $\mathsf{Z}_{<j-1},\Pi_{<j}$, for any fixed value of z the value of T_z is independent of the value of Z_{j-1}:

- If $j-1$ is odd, T_z is a function of (C,D) and Z_{j-1} is a function of (A,B).
- If $j-1$ is even, T_z is a function of (A,B) and Z_{j-1} is a function of (C,D).
- And, furthermore, (A,B) and (B,D) are independent given $\mathsf{Z}_{<j-1},\Pi_{<j}$.

[1] Intuitively, the statement says that, if π_{BMHPC} communicates little, then the distribution of Z_j, conditioned on the information available at the end of round j, is close to uniform.

Lemma 4.8 says that if the protocol were to reveal information about the solution T_z of a truly uniform instance z, then this would give us a protocol for ε-solving set disjointness. But, by induction, we know that the chosen instance $z \sim \mathsf{Z}_{j-1}$ is close to uniform. It should then follow from Proposition 3.9 that if the protocol reveals some information about this almost uniform instance, it must also reveal information about a truly uniform instance. Let us now make this precise.

Without loss of generality suppose that $j-1$ is odd and hence $z \in \mathcal{Y}$. Then we can upper bound the last expression above by:[2]

$$
\begin{aligned}
&\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\Big[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{Z}_j))\Big] \\
&\quad\leqslant \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j-1},\Pi_{<j})}\left[6\cdot \mathop{\mathbb{E}}_{(z\sim\mathcal{U}_\mathcal{Y})}\Big[\Lambda(\mu(\mathsf{T}^j_z \mid \mathsf{Z}_{<j-1},\Pi_{<j}),\mu(\mathsf{Z}_j))\Big]\right] \\
&\qquad\qquad + \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j-1},\Pi_{<j})}\Big[\Lambda(\mu(\mathsf{Z}_{j-1} \mid \mathsf{E}_{j-1},\Pi_{j-1}),\mathcal{U}_\mathcal{Y})\Big] \\
&\qquad\qquad\qquad\qquad \text{(by Proposition 3.9 and linearity of expectatiom)} \\
&\quad= 6\cdot \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j-1},\Pi_{<j})}\left[\mathop{\mathbb{E}}_{(z\sim\mathcal{U}_\mathcal{Y})}\Big[\Lambda(\mu(\mathsf{T}^j_z \mid \mathsf{Z}_{<j-1},\Pi_{<j}),\mu(\mathsf{Z}_j))\Big]\right] \\
&\qquad\qquad + \mathop{\mathbb{E}}_{(\mathsf{Z}_{<j-1},\Pi_{<j})}\Big[\Lambda(\mu(\mathsf{Z}_{j-1} \mid \mathsf{E}_{j-1},\Pi_{j-1}),\mu(\mathsf{Z}_{j-1}))\Big] \\
&\qquad\qquad\qquad\qquad \text{(by definition } z_{j-1}\in\mathcal{Y} \text{ and } \mu(\mathsf{Z}_{j-1})=\mathcal{U}_\mathcal{Y}.) \\
&\quad\leqslant \Bigg[6\cdot \underbrace{c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{B}) + \mathbb{I}(\Pi;\mathsf{B}^j \mid \mathsf{B}^{>j},\mathsf{A})}{m^2}}_{\text{Lemma 4.8}} \\
&\qquad + \underbrace{6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^{\leqslant j-1},\mathsf{C}^{\leqslant j-1} \mid \mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^{\leqslant j-1},\mathsf{D}^{\leqslant j-1} \mid \mathsf{B}^{>j-1},\mathsf{D}^{>j-1},\mathsf{A},\mathsf{C})}{m^2}}_{\text{induction hypothesis (case } j-1)}\Bigg] \\
&\quad= \Bigg[6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^j \mid \mathsf{A}^{<j},\mathsf{C}^{<j},\mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^j \mid \mathsf{B}^{>j},\mathsf{D}^{>j},\mathsf{A},\mathsf{C})}{m^2} \\
&\qquad + 6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^{\leqslant j-1},\mathsf{C}^{\leqslant j-1} \mid \mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^{\leqslant j-1},\mathsf{D}^{\leqslant j-1} \mid \mathsf{B}^{>j-1},\mathsf{D}^{>j-1},\mathsf{A},\mathsf{C})}{m^2}\Bigg] \\
&\quad\leqslant \Bigg[6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^j,\mathsf{C}^j \mid \mathsf{A}^{<j},\mathsf{C}^{<j},\mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^j,\mathsf{D}^j \mid \mathsf{B}^{>j},\mathsf{D}^{>j},\mathsf{A},\mathsf{C})}{m^2} \\
&\qquad + 6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^{\leqslant j-1},\mathsf{C}^{\leqslant j-1} \mid \mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^{\leqslant j-1},\mathsf{D}^{\leqslant j-1} \mid \mathsf{B}^{>j-1},\mathsf{D}^{>j-1},\mathsf{A},\mathsf{C})}{m^2}\Bigg] \\
&\quad= 6c\cdot\frac{\mathbb{I}(\Pi;\mathsf{A}^{\leqslant j},\mathsf{C}^{\leqslant j} \mid \mathsf{B},\mathsf{D}) + \mathbb{I}(\Pi;\mathsf{B}^{\leqslant j},\mathsf{D}^{\leqslant j} \mid \mathsf{B}^{>j},\mathsf{D}^{>j},\mathsf{A},\mathsf{C})}{m^2},
\end{aligned}
$$

where the last inequality follows from the data processing inequality (Fact E.2 Item 8). This concludes the proof. □

[2] In the first inequality it should be understood how Λ is playing two different roles. On the one hand it is being used as an information-theoretic quantity (the distance to uniform), and on the other it is being used in a capacity similar to how total variation distance appears in the obvious upper-bound $\mathbb{E}_\mu[f(x)] \leqslant \mathbb{E}_\nu[f(x)] + \|\mu-\nu\|_1 \cdot \max f(x)$.

4.4 ε-Solvers for Set-Int Give Exact Solvers

We now show a lower-bound against protocols trying to ε-solve Set-Intersection, by reducing from exactly solving Set-Intersection:

Theorem 4.12. *For any constant $\gamma \in (0,1)$, there exists a constant $c \geqslant 1$ such that, given a protocol $\pi_{\mathsf{Set\text{-}Int}}$ which internally ε-solves Set-Intersection (in Λ distance) on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ with $\varepsilon \geqslant \frac{8}{m}$, we may obtain a protocol π that exactly solves Set-Intersection on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$, such that $\mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi) \leqslant \frac{c}{\varepsilon} \cdot \mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) + \gamma m$.*

From the known fact (Proposition 3.2) that Set-Intersection has information complexity $\Omega(m)$ on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ we get the following:

Corollary 4.13. *There exists a constant $c > 0$ such that any protocol $\pi_{\mathsf{Set\text{-}Int}}$ that internally (Λ, ε)-solves SetInt_m on inputs from the distribution $\mathcal{D}_{\mathsf{Set\text{-}Int}}$, with $\varepsilon \geqslant 8/m$, has $\mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) \geqslant c \cdot \varepsilon m$.*

An analogue of Theorem 4.12 was proven in [AR20], where ε-solving was with respect to total variation distance instead of triangular discrimination. It was then shown that any protocol for ε-solving Set-Intersection in this TVD sense gives a protocol for Set-Intersection with a $O(1/\varepsilon^2)$ blowup in information cost. Theorem 4.12, instead, gives a blowup of $O(1/\varepsilon)$. So our assumption is stronger, but our conclusion is also stronger.

Proof of Theorem 4.12. We begin by recalling the notation we use for the Set-Intersection problem: Alice and Bob's input sets are denoted by X and Y respectively, and their corresponding random variables by X and Y. The unique element in the intersection $X \cap Y$ is denoted by e^*, and its random variable is e^*. Throughout this proof, for a permutation σ and a set S, $\sigma(S)$ denotes the set $\{\sigma(s) \mid s \in S\}$.
Without loss of generality, we may assume that we have a protocol $\pi_{\mathsf{Set\text{-}Int}}$ that internally ε-solves from *Alice*'s side Set-Intersection on inputs distributed according to $\mathcal{D}_{\mathsf{Set\text{-}Int}}$, that is,

$$\mathbf{\Lambda} = \mathop{\mathbb{E}}_{\mathsf{X},\mathsf{\Pi}}[\Lambda(\mu(\mathsf{e}^* \mid \mathsf{\Pi}, \mathsf{X}), \mu(\mathsf{e}^*, \mathsf{X}))] \geqslant \varepsilon,$$

where $\varepsilon \geqslant 8/m$. We will use $\pi_{\mathsf{Set\text{-}Int}}$ to design a protocol π which solves Set-Intersection on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$.

For now, suppose that Alice and Bob proceed as follows: they sample a uniformly random permutation Γ of $[m]$ using public randomness, and run the ε-solving protocol $\pi_{\mathsf{Set\text{-}Int}}$ on $(\Gamma(X), \Gamma(Y))$, obtaining a transcript Π. Alice now knows the induced distribution of the intersection point $\mu(\mathsf{\Gamma}(e^*) \mid \mathsf{\Pi} = \Pi, \mathsf{\Gamma}(X) = \Gamma(X))$, conditioned on the transcript she has seen, and Bob knows the analogous distribution with his input fixed. Without the conditioning on the transcript, we know by symmetry of $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ that $\mu(\mathsf{\Gamma}(e^*) \mid \mathsf{\Gamma}(X) = \Gamma(X))$ is uniform over $\Gamma(X)$. In fact, we have the following observation about the distribution of inputs generated by applying $\mathsf{\Gamma}$.

Observation 4.14. *For any fixed X and Y such that $|X| = |Y| = m/4$, and $|X \cap Y| = 1$, and a uniformly random permutation $\mathsf{\Gamma}$, the instance $(\mathsf{\Gamma}(X), \mathsf{\Gamma}(Y))$ is distributed according to $\mathcal{D}_{\mathsf{Set\text{-}Int}}$.*

To see this, consider the fixed values $X = \{1, \ldots m/4\}$ and $Y = \{m/4, \ldots m/2 - 1\}$. Because $\mathsf{\Gamma}$ is a uniformly random permutation, it maps $m/4$ to a uniformly random element of $[m]$, and the rest of X and Y to uniformly random subsets of size $m/4 - 1$. By symmetry, this argument applies to any fixed X and Y of size $m/4$ intersecting in exactly one element.

4.4.1 A single round of π

We start by describing a single round of the protocol π. On the input (X, Y):

1. Alice and Bob publicly sample a random permutation Γ of $[m]$, and apply it to X and Y to obtain $\Gamma(X)$ and $\Gamma(Y)$ respectively.
2. They run $\pi_{\mathsf{Set\text{-}Int}}$ on $\Gamma(X)$ and $\Gamma(Y)$ to obtain a transcript Π. By doing so, Alice learns a distribution $q_1^\Pi, \ldots, q_m^\Pi$, where
$$q_i^\Pi = \Pr[\mathsf{\Gamma}(e^*) = i \mid \mathsf{\Pi} = \Pi, \mathsf{\Gamma}(X) = \Gamma(X)].$$
3. Alice uses q^Π to assign "scores" to each element e of $[m]$, which reflect the likelihood that $\{e\}$ is the intersection of X and Y.

This strategy follows the general idea of [AR20], but where they assigned the score 1 to all e such that $q^\Pi_{\Gamma(e)}$ is in the largest-probability half of $q_1^\Pi, \ldots, q_m^\Pi$ (and this works for ε-solving under TVD), we assign scores in a smoother fashion. Let $i = \Gamma(e)$ and $n = m/4$ for brevity. Then if $q_i^\Pi \leqslant 1/n$, we give e a score of 0. Otherwise, its score is $(q_i^\Pi - 1/n)/(q_i^\Pi + 1/n)$. This choice of the "scoring function" may seem odd at this juncture, so we offer two comments about it:

- Note that this is equal to $1 - (2/n)/(q_i^\Pi + 1/n)$, which is an increasing function of q_i^Π, and hence passes a basic sanity test (i.e. if $\pi_{\mathsf{Set\text{-}Int}}$ says an element is more likely to be $\Gamma(e^*)$, we believe it).
- The scoring function was chosen to make the expected score of e^* larger than the expected score of any non-e^* element by at least $\mathbf{\Lambda}$, and the variance of every element's score turned out to be small.

We package the properties of this scoring function into a few claims below. Throughout these claims, we will use the shorthand notation $\mathsf{A} \mid B$ to mean $\mathsf{A} \mid \mathsf{B} = B$, where A and B are random variables, and B is some fixed value that B can take.

Claim 4.15. *For any non-e^* element $e \in [m]$,*
$$\mathop{\mathbb{E}}_{\mathsf{\Gamma}}[\mathsf{Score}(e^*) - \mathsf{Score}(e)] \geqslant \mathbf{\Lambda}/2.$$

Proof. The proof is in two parts; first we bound $\mathsf{Score}(e^*)$ under some appropriate conditioning, and then $\mathsf{Score}(e)$ for non-e^* elements.
Fix any (X, Y) with $|X| = |Y| = n$ and $|X \cap Y| = 1$. Consider a fixed choice of $\Gamma(X)$ and the transcript Π, and hence a distribution q^Π. Let P denote the set of indices where $q_i^\Pi > 1/n$ (i.e. those that lead to positive scores). Note that $P \subseteq \Gamma(X)$; because we condition on $\Gamma(X)$ (and more) to obtain q^Π. By definition, the probability that e^* is mapped to i is q_i, and hence:
$$\mathop{\mathbb{E}}_{\mathsf{\Gamma} \mid \Gamma(X), \Pi}[\mathsf{Score}(e^*)] = \sum_{i \in P} q_i^\Pi \cdot \frac{q_i^\Pi - 1/n}{q_i^\Pi + 1/n}.$$
On the other hand, fixing both $\Gamma(X)$, $\Gamma(Y)$ (and hence also $\Gamma(e^*)$) and Π, fixes the distribution q^Π, but leaves completely free the map $\Gamma : X \setminus \{e^*\} \to \Gamma(X) \setminus \{\Gamma(e^*)\}$. So we find that:
$\mathsf{\Gamma}(e)$ is uniformly random over $\Gamma(X) \setminus \{\Gamma(e^*)\}$ for any non-e^* element $e \in X$.
This means that for any $i \in [m]$,
$$\Pr_{\mathsf{\Gamma} \mid \Gamma(X), \Gamma(Y), \Pi}[\mathsf{\Gamma}(e) = i] \leqslant \frac{1}{n-1}.$$

Hence we can compute the expected score of e as:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\Gamma(X),\Gamma(Y),\Pi}[\mathsf{Score}(e)] &= \sum_{i\in P}\Pr[\mathsf{\Gamma}(e)=i]\cdot\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n} \leqslant \sum_{i\in P}\frac{1}{n-1}\cdot\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n}\\ &= \sum_{i\in P}\left(\frac{1}{n}+\frac{1}{n(n-1)}\right)\cdot\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n} \leqslant \frac{1}{n}+\sum_{i\in P}\frac{1}{n}\cdot\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n},\end{aligned}$$

where the last inequality follows because $|P|\leqslant n-1$. And now taking expectation over $\Gamma(X)$ and Π:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\mathsf{\Pi},\mathsf{\Gamma}(X)}[\mathsf{Score}(e^*)-\mathsf{Score}(e)] &= \mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\left[\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\mathsf{\Pi},\mathsf{\Gamma}(X)}[\mathsf{Score}(e^*)]-\mathop{\mathbb{E}}_{\mathsf{\Gamma}(Y)|\mathsf{\Pi},\mathsf{\Gamma}(X)}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\mathsf{\Gamma}(X),\mathsf{\Gamma}(Y)}[\mathsf{Score}(e)]\right]\\ &\geqslant \mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\left[\sum_{i\in P}\frac{\left(q_i^\Pi-1/n\right)^2}{q_i^\Pi+1/n}-1/n\right]\\ &= \mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\left[\Lambda(q^\Pi,\mathcal{U}_{\Gamma(X)})\right]-1/n = \mathbf{\Lambda}-1/n \geqslant \mathbf{\Lambda}/2.\end{aligned}$$

where the last inequality is because $\mathbf{\Lambda}\geqslant\varepsilon\geqslant 8/m = 2/n$.

□

Claim 4.16. *The variance of* $\mathsf{Score}(e^*)$ *is at most* $\mathbf{\Lambda}$.

Proof. Recall that $\mathrm{Var}[X]\leqslant\mathbb{E}[X^2]$ for any random variable X. As before, we will bound $\mathbb{E}[\mathsf{Score}(e^*)]$ by conditioning on fixed choices of $\Gamma(X)$ and the transcript Π, which gives Alice the distribution q^Π. Also as before, let P denote the set of indices i such that $q_i\geqslant 1/n$.

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\Gamma(X),\Pi}\left[\mathsf{Score}(e^*)^2\right] &= \sum_{i\in P}q_i^\Pi\cdot\left(\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n}\right)^2\\ &= \sum_{i\in P}\frac{q_i^\Pi}{q_i^\Pi+1/n}\cdot\frac{(q_i^\Pi-1/n)^2}{q_i^\Pi+1/n}\\ &\leqslant \Lambda(q^\Pi,\mathcal{U}_{\Gamma(X)}).\end{aligned}$$

And taking expectation over $\Gamma(X)$ and Π:

$$\mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\mathsf{\Gamma}(X),\mathsf{\Pi}}\left[\mathsf{Score}(e^*)^2\right] \leqslant \mathop{\mathbb{E}}_{\mathsf{\Gamma}(X),\mathsf{\Pi}}\left[\Lambda(q^\Pi,\mathcal{U}_{\mathsf{\Gamma}(X)})\right] = \mathbf{\Lambda}.$$

□

Claim 4.17. *For any* $e(\neq e^*)\in X$, *the variance of* $\mathsf{Score}(e)$ *is at most* $2\mathbf{\Lambda}$.

Proof. Again, because $\mathrm{Var}[X]\leqslant\mathbb{E}[X^2]$ for any random variable X, and conditioning on fixed choices of $\Gamma(X),\Gamma(Y)$ and Π, $\mathsf{\Gamma}(e)$ for $e\in X\setminus\{e^*\}$ is uniformly random over $\Gamma(X\setminus\{e^*\})$:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathsf{\Gamma}|\Gamma(X),\Gamma(Y),\Pi}\left[\mathsf{Score}(e)^2\right] &= \sum_{i\in P}\frac{1}{n-1}\cdot\left(\frac{q_i^\Pi-1/n}{q_i^\Pi+1/n}\right)^2\\ &\leqslant \sum_{i\in P}\frac{1}{n-1}\cdot\frac{\left(q_i^\Pi-1/n\right)^2}{q_i^\Pi+1/n}\cdot\frac{1}{1/n}\\ &= \Lambda\big(q^\Pi,\mathcal{U}_{\Gamma(X)}\big)\cdot\left(1+\frac{1}{n-1}\right)\leqslant 2\Lambda\big(q^\Pi,\mathcal{U}_{\Gamma(X)}\big),\end{aligned}$$

where the first inequality holds because $q_i^{\Pi} \geqslant 0$. And taking expectation over $\Gamma(X), \Gamma(Y)$ and Π:

$$\mathop{\mathbb{E}}_{\Gamma(X),\Pi}\left[\mathop{\mathbb{E}}_{\Gamma(Y)|\Gamma(X),\Pi}\left[\mathop{\mathbb{E}}_{\Gamma|\Gamma(X),\Gamma(Y),\Pi}\left[\mathsf{Score}(e)^2\right]\right]\right] \leqslant \mathop{\mathbb{E}}_{\Gamma(X),\Pi}\left[2\Lambda\left(q^{\Pi}, \mathcal{U}_{\Gamma(X)}\right)\right] = 2\boldsymbol{\Lambda}.$$

□

4.4.2 The actual protocol

We are ready to describe the protocol π. Let $\gamma \in (0,1)$ be any constant. We will run $k = 1600/(\varepsilon\gamma^2)$ (independent) rounds of the protocol in Section 4.4.1, and for each element $e \in X$, Alice will add up its scores in each round to obtain the total score $\mathsf{TScore}(e)$. Claim 4.15 tells us that we should expect the total score for e^* to be higher than the total score of an element $e \neq e^*$. We then choose the threshold

$$\tau = \left(\mathbb{E}[\mathsf{TScore}(e^*)] + \mathbb{E}[\mathsf{TScore}(e)]\right)/2,$$

to be exactly in the middle of these two expected values.

Alice will then send to Bob all elements in X which have TScore at least τ; let us call this set S. Bob will then compute the intersection $S \cap Y$, and report the unique element that is (hopefully) present there as the answer.

To complete the reduction, we need to show that:

- The total score of the target element $\mathsf{TScore}(e^*)$ is at least τ. This will ensure that $e^* \in S$, which means that π is correct.
- The number of elements Alice sends is *small*. This is required to control the information cost of π.

We will get both of these properties only with some (constant) high probability; in particular, if S is too large, Alice will just report failure instead of communicating S.

Claim 4.18. *$\mathsf{TScore}(e^*) \leqslant \tau$ with probability at most $1/100$.*

Proof. Using Chebyshev's inequality on the total score of e^*, we have:

$$\begin{aligned}
&\Pr\left[\left|\mathsf{TScore}(e^*) - \mathbb{E}[\mathsf{TScore}(e^*)]\right| > 1/2 \cdot \mathbb{E}[\mathsf{TScore}(e^*) - \mathsf{TScore}(e)]\right] \\
&\quad\leqslant \frac{\mathrm{Var}[\mathsf{TScore}(e^*)]}{(1/2 \cdot \mathbb{E}[\mathsf{TScore}(e^*) - \mathsf{TScore}(e)])^2} = \frac{k \cdot \mathrm{Var}[\mathsf{Score}(e^*)]}{k^2/4 \cdot \mathbb{E}[\mathsf{Score}(e^*) - \mathsf{Score}(e)]^2} \\
&\quad\leqslant \frac{4\boldsymbol{\Lambda}}{k \cdot (\boldsymbol{\Lambda}/2)^2} \leqslant 16/k\varepsilon = \gamma^2/100,
\end{aligned}$$

where the second inequality follows from Claims 4.15 and 4.16. Hence the probability that $\mathsf{TScore}(e^*)$ is less than τ is at most $1/100$. □

Claim 4.19. *For any non-e^* element $e \in X$, $\mathsf{TScore}(e) \geqslant \tau$ with probability at most $\gamma^2/50$.*

Proof. Using Chebyshev on the total score of $e \neq e^*$, we have:

$$\begin{aligned}
&\Pr\Big[\Big|\mathsf{TScore}(e) - \mathbb{E}[\mathsf{TScore}(e)]\Big| > 1/2 \cdot \mathbb{E}[\mathsf{TScore}(e^*) - \mathsf{TScore}(e)]\Big] \\
&\quad\leqslant \frac{\mathrm{Var}[\mathsf{TScore}(e)]}{(1/2 \cdot \mathbb{E}[\mathsf{TScore}(e^*) - \mathsf{TScore}(e)])^2} = \frac{k \cdot \mathrm{Var}[\mathsf{Score}(e)]}{k^2/4 \cdot \mathbb{E}[\mathsf{Score}(e^*) - \mathsf{Score}(e)]^2} \\
&\quad\leqslant \frac{4 \cdot 2\boldsymbol{\Lambda}}{k \cdot (\boldsymbol{\Lambda}/2)^2} \leqslant 32/k\varepsilon = \gamma^2/50,
\end{aligned}$$

where the second inequality follows from Claims 4.15 and 4.17. □

This means that the expected size of $S \setminus \{e^*\}$ is at most $\gamma^2 m/50$, and hence by Markov's this number is $> \gamma^2 m/10$ with probability $\leqslant 1/5$. If the number of elements above the threshold exceeds $\gamma^2 m/10 + 1$, then the protocol π fails.

And now we have everything we need to bound the information cost of π. Note that this part of the argument is exactly as in [AR20]. By the chain rule of mutual information, the information cost of π on $\mathcal{D}_{\mathsf{Set\text{-}Int}}$ can be broken down into three components:

- The mutual information $\mathbb{I}(\Gamma_1, \ldots, \Gamma_k, R_1, \ldots, R_k; \mathsf{X} \mid \mathsf{Y})$ and the symmetric term with X and Y exchanging roles, where Γ_i denotes the random permutations used to scramble the input in the $i-$th round of π, and R_i denotes the randomness of the i-th run of $\pi_{\mathsf{Set\text{-}Int}}$. These are both 0, because the permutations and the randomness are chosen independently of the inputs X and Y.

- The mutual information $\mathbb{I}(\Pi_1, \ldots, \Pi_k; \mathsf{X} \mid \mathsf{Y}, \Sigma, \mathsf{R})$ (and the symmetric term with X and Y swapped), where $\Pi_1, \ldots, \Pi_k$ denote the transcripts of the k runs of $\pi_{\mathsf{Set\text{-}Int}}$, $\Sigma = \Gamma_1, \ldots, \Gamma_k$, and $\mathsf{R} = R_1, \ldots, R_k$. By the chain rule of mutual information, we have:

  $$\mathbb{I}(\Pi_1, \ldots, \Pi_k; \mathsf{X} \mid \mathsf{Y}, \Sigma, \mathsf{R}) \leqslant \sum_{i=1}^{k} \mathbb{I}(\Pi_i; \mathsf{X} \mid \mathsf{Y}, \Sigma, \mathsf{R}, \Pi_{<i}).$$

  Observe that conditioning on X, Y, Γ_i, and R_i fixes Π_i, and hence makes it independent of $\Pi_{<i}$, Γ_{-i}, and R_{-i}. Applying Proposition E.4, we get that the i-th term of the sum above is upper bounded by $\mathbb{I}(\Pi_i; \mathsf{X} \mid \mathsf{Y}, \Gamma_i, R_i)$. But now (once again) Π_i is fixed after conditioning on $\Gamma_i(\mathsf{X})$, $\Gamma_i(\mathsf{Y})$, and R_i, and hence independent of the remaining parts of Γ_i and X and Y. So by Proposition E.4, each term in the sum is bounded above by $\mathbb{I}(\Pi_i; \Gamma_i(\mathsf{X}) \,|\, \Gamma(\mathsf{Y}), R_i)$. Repeating this entire argument for the symmetric term tells us that:

  $$\mathbb{I}(\Pi_1, \ldots, \Pi_k; \mathsf{X}|\mathsf{Y}, \Sigma, \mathsf{R}) + \mathbb{I}(\Pi_1, \ldots, \Pi_k; \mathsf{Y}|\mathsf{X}, \Sigma, \mathsf{R}) \leqslant \sum_{i=1}^{k} \mathbb{I}(\Pi_i; \Gamma_i(\mathsf{X})|\Gamma(\mathsf{Y}), R_i) + \mathbb{I}(\Pi_i; \Gamma_i(\mathsf{Y})|\Gamma(\mathsf{X}), R_i),$$

  which is exactly $k \cdot \mathsf{IC}^{\mathrm{int}}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}})$.

- The mutual information between the final set S Alice sends and her input. This is trivially upper bounded by $\mathbb{H}(\mathsf{S})$, which (since S is a set of size at most $\gamma^2 \cdot m/10$) is upper bounded by $H_2(\gamma^2/10) \cdot m$. Since $H_2(p) \leqslant 2\sqrt{p \cdot (1-p)} \leqslant 2\sqrt{p}$ (by Fact E.1), this is at most γm.

So letting $c = \frac{1600}{\gamma^2}$, we get that $\mathsf{IC}^{\mathrm{int}}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi) \leqslant \frac{c}{\varepsilon} \mathsf{IC}^{\mathrm{int}}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) + \gamma m$. □

Remark 4.20. The protocol π actually works in the worst case, that is, it needs no assumptions about X and Y other than $|X| = |Y| = m/4$, and that $|X \cap Y| = 1$. But of course without a distribution on its inputs we cannot talk about its information cost.

4.5 Short Protocols for BHPC Give Low-Information ε-Solvers for Set-Int

We also wish to prove a tight lower bound for $\mathsf{BHPC}_{m,r}$. Note that this lower-bound is not necessary to derive our lower-bounds for k-cores and degeneracy, so this section may be skipped for that purpose. The lower-bound ultimately follows from the following lemma, which is analogous to Lemma 4.11.

Lemma 4.21. *There is a constant $c > 0$ such that the following statement holds for all $j \in [r]$:*[3]

$$\mathsf{Hypothesis}_j: \quad \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} [\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{Z}_j))] \leqslant j \cdot c \cdot \left(\frac{CC(\pi_{\mathsf{BHPC}}) + r \log m}{m^2} + \frac{r}{m} \right).$$

One can already see that the statement is more complicated. This is owed to the possibility of the pointers pointing either to some of the (few) instances about which the protocol has revealed information, and the possibility of the pointers looping. These two obstacles raise technical difficulties, which are overcome in two ways: the number of jumps r is assumed to be smaller than $\sqrt{m}$, so looping is unlikely, and there are only $O(m/r)$ instances about which the protocol could have revealed significant information, hence when hiding an instance of set-intersection in a random position, as we did before, it is unlikely we will hit such an instance.

As before, this is proven by induction. The base case $\mathsf{Hypothesis}_1$ is easy to see: note that $\mu(\mathsf{Z}_1 \mid \mathsf{E}_1, \Pi_1) = \mu(\mathsf{Z}_1)$ as Z_1 is independent of $\mathsf{E}_1 = z_0$ which is fixed and Z_1 is independent of Π_1 as Π_1 is a function if (C, D). We obtain $\mathsf{Hypothesis}_j$ from $\mathsf{Hypothesis}_{j-1}$ in two steps: we first show that if the protocol reveals a lot of information about the variable Z_j, then it must essentially be "almost-solving" a *uniformly random instance*. This is because $Z_j = T_{Z_{j-1}}$ and Z_{j-1} is, under $\mathsf{Hypothesis}_{j-1}$, close to uniform. We then show that this is impossible for a short protocol to do, namely:

Lemma 4.22. *There is a constant $c' > 0$ such that for any $j \in [r]$, we have:*

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x \sim \mathcal{U}_{\mathcal{X}}} [\Lambda(\mu(\mathsf{T}_x \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_x))] \leqslant c' \cdot \left(\frac{CC(\pi_{\mathsf{BHPC}}) + r \log m}{m^2} + \frac{j}{m} \right)$$

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{y \sim \mathcal{U}_{\mathcal{Y}}} [\Lambda(\mu(\mathsf{T}_y \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_y))] \leqslant c' \cdot \left(\frac{CC(\pi_{\mathsf{BHPC}}) + r \log m}{m^2} + \frac{j}{m} \right).$$

We state the proof of Lemma 4.22 shortly. Before that, we show how Lemma 4.22 lets us complete the induction in order to prove Lemma 4.21. This also follows very closely to the analogous proof of [ACK19] where they showed it for total variation distance.

Proof of Lemma 4.21. The information—measured by Λ-distance to uniform—that the j-th round of the protocol reveals about $\mathsf{Z}_j = \mathsf{T}_{\mathsf{Z}_{j-1}}$ is the information revealed about an instance sampled

[3] Intuitively, the statement says that, if π_{BHPC} communicates little, then the distribution of Z_j, conditioned on the information available at the end of round j, is close to uniform.

according to Z_{j-1}:

$$\begin{aligned}
\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\Big[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{Z}_j))\Big] &= \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\Big[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j),\mu(\mathsf{Z}_j))\Big] \\
&= \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j},\Pi^{<j})}\Big[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{Z}^{<j},\Pi^{<j}),\mu(\mathsf{Z}_j))\Big] \\
&\qquad \text{(by definition of } \mathsf{E}_j := (\mathsf{Z}^{<j},\Pi^{<j}), \text{ and since } \mathsf{Z}_j \text{ and } \Pi_j \text{ are independent given } \mathsf{E}_j.) \\
&= \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j},\Pi^{<j})}\Big[\Lambda(\mu(\mathsf{T}_{\mathsf{Z}_{j-1}} \mid \mathsf{Z}^{<j-1},\mathsf{Z}_{j-1},\Pi^{<j}),\mu(\mathsf{Z}_j))\Big] \\
&\qquad \text{(by definition, the pointer } Z_j = T_{Z_{j-1}}) \\
&= \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j-1},\Pi^{<j})}\ \mathop{\mathbb{E}}_{z\sim\mathsf{Z}_{j-1}|(\mathsf{Z}^{<j-1},\Pi^{<j})}\Big[\Lambda(\mu(\mathsf{T}_z \mid \mathsf{Z}^{<j-1},\Pi^{<j}),\mu(\mathsf{Z}_j))\Big].
\end{aligned}$$

We removed the conditioning on Z_{j-1} in the first distribution because, conditioned on knowing $\mathsf{Z}^{<j-1},\Pi^{<j}$, for any fixed value of z the value of T_z is independent of the value of Z_{j-1}:

- If $j-1$ is odd, T_z is a function of (C,D) and Z_{j-1} is a function of (A,B).
- If $j-1$ is even, T_z is a function of (A,B) and Z_{j-1} is a function of $(\boldsymbol{C},\boldsymbol{D})$.
- And, furthermore, (A,B) and (B,D) are independent given $\mathsf{Z}^{<j-1},\Pi^{<j}$.

Lemma 4.22 says that if the protocol were to reveal information about the solution T_z of a truly uniform instance z, then this would give us a protocol for ε-solving set disjointness. But, by induction, we know that the chosen instance $z \sim \mathsf{Z}_{j-1}$ is close to uniform. It should then follow from Proposition 3.9 that if the protocol reveals some information about this almost uniform instance, it must also reveal information about a truly uniform instance. Let us now make this precise.

Without loss of generality suppose that $j-1$ is odd and hence $z \in \mathcal{Y}$. Then we can upper bound the last expression above by:

$$\begin{aligned}
&\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)}\Big[\Lambda(\mu(\mathsf{Z}_j \mid \mathsf{E}_j,\Pi_j),\mu(\mathsf{Z}_j))\Big] \\
&\quad\leqslant \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j-1},\Pi^{<j})}\left[6\cdot\mathop{\mathbb{E}}_{(z\sim\mathcal{U}_\mathcal{Y})}\Big[\Lambda(\mu(\mathsf{T}_z \mid \mathsf{Z}^{<j-1},\Pi^{<j}),\mu(\mathsf{Z}_j))\Big]\right] \\
&\qquad\qquad + \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j-1},\Pi^{<j})}\Big[\Lambda(\mu(\mathsf{Z}_{j-1} \mid \mathsf{E}_{j-1},\Pi_{j-1}),\mathcal{U}_\mathcal{Y})\Big] \\
&\qquad\qquad \text{(by Proposition 3.9 and linearity of expectatiom)} \\
&\quad= 6\cdot\mathop{\mathbb{E}}_{(\mathsf{Z}^{<j-1},\Pi^{<j})}\left[\mathop{\mathbb{E}}_{(z\sim\mathcal{U}_\mathcal{Y})}\Big[\Lambda(\mu(\mathsf{T}_z \mid \mathsf{Z}^{<j-1},\Pi^{<j}),\mu(\mathsf{Z}_j))\Big]\right] \qquad (*) \\
&\qquad\qquad + \mathop{\mathbb{E}}_{(\mathsf{Z}^{<j-1},\Pi^{<j})}\Big[\Lambda(\mu(\mathsf{Z}_{j-1} \mid \mathsf{E}_{j-1},\Pi_{j-1}),\mu(\mathsf{Z}_{j-1}))\Big] \\
&\qquad\qquad \text{(by definition } z_{j-1} \in \mathcal{Y} \text{ and } \mu(\mathsf{Z}_{j-1}) = \mathcal{U}_\mathcal{Y}.) \\
&\quad\leqslant 6\cdot\underbrace{c'\cdot\left(\frac{CC(\pi_{\mathsf{BHPC}})+r\log m}{m^2}+\frac{j}{m}\right)}_{\text{Lemma 4.22}} + \underbrace{(j-1)\cdot c\cdot\left(\frac{CC(\pi_{\mathsf{BHPC}})+r\log m}{m^2}+\frac{r}{m}\right)}_{\mathsf{Hypothesis}_{j-1}} \\
&\quad\leqslant j\cdot c\cdot\left(\frac{CC(\pi_{\mathsf{BHPC}})+r\log m}{m^2}+\frac{r}{m}\right) \\
&\qquad\qquad \text{(where we replaced } j\leqslant r \text{ by } r \text{ in the first term and set } c = 6c')
\end{aligned}$$

This concludes the proof. □

We now prove Lemma 4.22. The proof shows that any protocol that reveals information about the solution T_z to random instance z can be converted to a low-information protocol for externally ε-solving Set-Intersection. This will later be shown to be impossible. Let us state the lemma again:

Lemma 4.22. *There is a constant $c' > 0$ such that for any $j \in [r]$, we have:*

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_\mathcal{X}} [\Lambda(\mu(\mathsf{T}_x \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_x))] \leqslant c' \cdot \left(\frac{CC(\pi_{\mathsf{BHPC}}) + r\log m}{m^2} + \frac{j}{m}\right)$$
$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{y\sim\mathcal{U}_\mathcal{Y}} [\Lambda(\mu(\mathsf{T}_y \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_y))] \leqslant c' \cdot \left(\frac{CC(\pi_{\mathsf{BHPC}}) + r\log m}{m^2} + \frac{j}{m}\right).$$

Suppose towards a contradiction that this equation does not hold. We use π_{HPC} to design a protocol π_{SI} that can externally ε-solve the Set-Intersection problem (A_x, B_x) for a uniformly at random chosen $x \in \mathcal{X}$ and appropriately chosen $\varepsilon \in (0,1)$ to be determined later (see Definition 4.3 for the notion of ε-solving).

Protocol π_{SI}: The protocol for ε-solving Set-Int using a protocol π_{HPC} for $\mathsf{BHPC}_{m,r}$.
Input: An instance $(X, Y) \sim \mathcal{D}_{\text{Set-Int}}$ over the universe $\mathcal{Y}$ and a number $j \in [r]$.

1. **Sampling the instance.** Alice and Bob create an instance (A, B, C, D) of $\mathsf{BHPC}_{m,r}$ as follows:
   (a) Using public coins, Alice and Bob sample an index $i \in [m]$ uniformly at random, and Alice sets $A_{x_i} = X$ and Bob sets $B_{x_i} = Y$ using their given inputs in Set-Int.
   (b) Using public coins, Alice and Bob sample $A_{x_1}, \ldots, A_{x_{i-1}}$ from the A-side marginal of $\mathcal{D}_{\text{Set-Int}}$, and they sample $B_{x_{i+1}}, \ldots, B_{x_m}$ from the B-side marginal of $\mathcal{D}_{\text{Set-Int}}$.
   (c) Using private coins, Alice samples A_{x_r} for $r \in \{i+1, \ldots, n\}$ so that $(A_{x_r}, B_{x_r}) \sim \mathcal{D}_{\text{Set-Int}}$. Similarly Bob samples B_{x_ℓ} for $\ell \in \{1, \ldots, i-1\}$. This completes construction of $(\boldsymbol{A}, \boldsymbol{B})$.
   (d) Using public coins, Alice and Bob sample (C, D) completely from $\mathcal{D}_{\mathsf{BHPC}}$ (this is possible by Observation 4.6 as $(\mathsf{A}, \mathsf{B}) \perp (\mathsf{C}, \mathsf{D})$).
2. **Computing the answer.** For $q = 1$ to j, Alice and Bob do the following:
   (a) **Running protocol:** In round q, Alice and Bob run the round q of the protocol π_{BHPC} on the instance (A, B, C, D) by Alice playing P_A, Bob playing P_B, and both Alice and Bob simulating P_C and P_D with no communication (this is possible as both Alice and Bob know (C, D) entirely). Denote this transcript by Π_q.
   (b) **Fixing pointer:** Alice and Bob compute the pointer z_q using the fact that for any underlying instance $(A_x, B_x) \in (A, B) \setminus (A_{x_i}, B_{x_i})$ either Alice or Bob knows the entire instance. We use Π^* to denote the cumulative transcript generated by this step in all rounds.
   (c) They stop if $x_i = z_q$.
3. The players return $\Pi_{\text{Set-Int}} := (\Pi_1, \ldots, \Pi_j, \Pi^*)$ from which $\mu(e^*)$ is calculated.

First, we show that the internal information cost of this protocol is small if the communication cost of Π_{BHPC} is small. This uses standard direct sum argument of information complexity. More concretely, we have the following:

Claim 4.23. $\mathsf{IC}_{\mathcal{D}}(\pi_{\mathsf{Set\text{-}Int}}) = O\left(\frac{CC(\pi_{\mathsf{BHPC}})}{m} + \frac{j \log m}{m}\right)$.

The proof of Claim 4.23 is similar to that of Claim 5.6 of [ACK19]. We defer it to the Section C. At this point, we also need a bound on how well $\Pi_{\mathsf{Set\text{-}Int}}$ can "almost" solve Set-Intersection. To this end, we have the following guarantee.

Claim 4.24. $\pi_{\mathsf{Set\text{-}Int}}$ *externally* ε*-solves Set-Intersection on* $\mathcal{D}$ *where:*

$$\varepsilon \geqslant \mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_{\mathcal{X}}} [\Lambda(\mu(\mathsf{T}_x \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_x))] - \frac{j}{m}.$$

Proof. This proof follows the proof of Claim 5.5 of [ACK19] almost verbatim. The only change is that we are working with positive triangular discrimination (Λ) instead of the total variation distance (Δ_{TVD}). However, as Λ is bounded by Δ_{TVD} (Proposition 3.8), a similar calculation yields the claim. By Definition 4.3, $\varepsilon = \mathbb{E}_{\Pi_{\mathsf{Set\text{-}Int}}}[\Lambda(\mu(\mathsf{T} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}))]$. So we have:

$$\begin{aligned}
\mathop{\mathbb{E}}_{\Pi_{\mathsf{Set\text{-}Int}}} [\Lambda(\mu(\mathsf{T} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}))] &= \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j,\Pi^*,\mathsf{I}} [\Lambda(\mu(\mathsf{T}_{x_\mathsf{I}} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}_{x_\mathsf{I}}))] && (\text{as } \mathsf{T} = \mathsf{T}_{x_\mathsf{I}}) \\
&= \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j} \mathop{\mathbb{E}}_{\mathsf{I}} \mathop{\mathbb{E}}_{\Pi^*|\mathsf{E}_j,\Pi_j,\mathsf{I}} [\Lambda(\mu(\mathsf{T}_{x_\mathsf{I}} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}_{x_i}))] && (\text{as } \mathsf{I} \text{ is independent of } (\mathsf{E}_j, \Pi_j)) \\
&= \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j} \left[\sum_i \frac{1}{m} \mathop{\mathbb{E}}_{\Pi^*|\mathsf{E}_j,\Pi_j,\mathsf{I}=i} [\Lambda(\mu(\mathsf{T}_{x_i} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}_{x_i}))] \right] && (\text{as } \mathsf{I} \sim \mathcal{U}_{[m]}) \\
&= \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j} \left[\sum_{x_i \in \mathsf{Z}^{<j}} \frac{1}{m} \Lambda(\mu(\mathsf{T}_{x_i} \mid \Pi_{\mathsf{Set\text{-}Int}}), \mu(\mathsf{T}_{x_i})) + \sum_{x_i \notin \mathsf{Z}^{<j}} \frac{1}{m} \Lambda(\mu(\mathsf{T}_{x_i} \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_{x_i})) \right] \\
& && (\text{If } x_i \notin \mathsf{Z}^{<j} \text{ then } \Pi_{\mathsf{Set\text{-}Int}} = (\mathsf{E}_j, \Pi_j)) \\
&\geqslant \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j} \left[\sum_{x_i \notin \mathsf{Z}^{<j}} \frac{1}{m} \Lambda(\mu(\mathsf{T}_{x_i} \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_{x_i})) \right] \\
& && (\text{as the first sum term is at least 0}) \\
&\geqslant \mathop{\mathbb{E}}_{\mathsf{E}_j,\Pi_j} \mathop{\mathbb{E}}_{i} [\Lambda(\mu(\mathsf{T}_{x_i} \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_{x_i}))] - \frac{j}{m}. && (\text{as } \Lambda \text{ is upper bounded by 1})
\end{aligned}$$

□

Using Claim 4.23 and the lower bound on $\mathsf{IC}_{\mathcal{D}}(\pi_{\mathsf{Set\text{-}Int}})$ (Corollary 4.13), we have:

$$\varepsilon = O\left(\frac{CC(\pi_{\mathsf{BHPC}})}{m^2} + \frac{j \log m}{m^2}\right)$$

Now, using Claim 4.24, we get:

$$\mathop{\mathbb{E}}_{(\mathsf{E}_j,\Pi_j)} \mathop{\mathbb{E}}_{x\sim\mathcal{U}_{\mathcal{X}}} [\Lambda(\mu(\mathsf{T}_x \mid \mathsf{E}_j, \Pi_j), \mu(\mathsf{T}_x))] - \frac{j}{n} = O\left(\frac{CC(\pi_{\mathsf{BHPC}})}{m^2} + \frac{j \log m}{m^2}\right),$$

which concludes the proof of Lemma 4.22.

4.6 Putting it All Together

Given Lemma 4.11, we prove Theorem 4.1 in the following way: Consider a protocol π that is solving $\mathsf{BMHPC}_{m,r}$ with communication cost $o(m^2)$. By Lemma 4.11, we have:

$$\begin{aligned}\mathop{\mathbb{E}}_{(\mathsf{E}_r,\Pi_r)}\left[\Lambda(\mu(\mathsf{Z}_r \mid \mathsf{E}_r,\Pi_r),\mu(\mathsf{Z}_r))\right] &\leqslant 6c\cdot\frac{\mathbb{I}(\Pi\,;\mathsf{A},\mathsf{C}\mid\mathsf{B},\mathsf{D})+\mathbb{I}(\Pi\,;\mathsf{B},\mathsf{D}\mid\mathsf{A},\mathsf{C})}{m^2}\\ &\mathop{\leqslant}_{(*)} 12c\cdot\frac{\mathsf{CC}(\pi_{\mathsf{BHPC}})}{m^2}=o(1),\end{aligned}$$

where inequality $(*)$ follows from Items 1 and 3 in Fact E.2. However, next, we show that, for a correct protocol for $\mathsf{BMHPC}_{m,r}$, this quantity should be $\Omega(1)$, which contradicts the assumption that $\mathsf{CC}(\pi_{\mathsf{BHPC}})=o(m^2)$. An entirely similar argument derives Theorem 4.2 from Lemma 4.21. The only difference between these two arguments is the bound on r: We have to assume $r=o(\sqrt{m})$ because of the additive r^2/m factor in Lemma 4.21 (when applied with on the r-th round of the protocol).

Lemma 4.25. *Any protocol that solves $\mathsf{BMHPC}_{m,r}$ on distribution $\mathcal{D}_{\mathsf{BMHPC}}$, or $\mathsf{BHPC}_{m,r}$, on distribution $\mathcal{D}_{\mathsf{BHPC}}$, with constant probability must have:*

$$\mathop{\mathbb{E}}_{(\mathsf{E}_r,\Pi_r)}\left[\Lambda(\mu(\mathsf{Z}_r \mid \mathsf{E}_r,\Pi_r),\mu(\mathsf{Z}_r))\right]=\Omega(1).$$

Proof. Let b denote the modulus function such that $b(z_k)=i \mod 2$ iff $z_r=x_i$ or y_i depending on where r is even or odd. If the protocol π computes $\mathsf{BMHPC}_{m,r}$ or $\mathsf{BHPC}_{m,r}$ with probability $2/3$, we have the following inequality:

$$\begin{aligned}2/3&=\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}\Pr_{\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r}[b(\mathsf{Z}_r)=\mathcal{O}_\pi(\mathsf{E}_r,\Pi_r)]\\ &\mathop{\leqslant}_{(1)}\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}\left[\Pr_{\mathsf{Z}_r\sim\mathcal{U}_\mathcal{X}}[b(\mathsf{Z}_r)=\mathcal{O}_\pi(\mathsf{E}_r,\Pi_r)]+\|\mu(b(\mathsf{Z}_r)\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_{0,1}\|_1\right]\\ &\mathop{\leqslant}_{(2)}\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}\left[\Pr_{\mathsf{Z}_r\sim\mathcal{U}_\mathcal{X}}[b(\mathsf{Z}_r)=\mathcal{O}_\pi(\mathsf{E}_r,\Pi_r)]+\|\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_\mathcal{X}\|_1\right]\\ &=\frac{1}{2}+\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}[\|\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_\mathcal{X}\|_1]\end{aligned}$$

where (1) holds because of the simple fact that $\Pr_p[\mathcal{E}]\leqslant\Pr_q[\mathcal{E}]+\|p-q\|_1$ where p and q are two distributions over a sample space of which $\mathcal{E}$ is an event (see Footnote 2), and (2) holds because of data processing inequality. This implies:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}[\|\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_{0,1}\|_1]\geqslant 1/6 &\Rightarrow\left(\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}[\|\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_{0,1}\|_1]\right)^2\geqslant 1/36\\ &\Rightarrow\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}\left[\|\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r)-\mathcal{U}_{0,1}\|_1^2\right]\geqslant 1/36\\ &\Rightarrow\mathop{\mathbb{E}}_{\mathsf{E}_r,\Pi_r}[\Lambda(\mu(\mathsf{Z}_r\mid\mathsf{E}_r,\Pi_r),\mathcal{U}_{0,1})]\geqslant 1/288=\Omega(1),\end{aligned}$$

which completes the proof. □

5 Reduction From BMHPC to Degeneracy

In this section we prove streaming lower bounds for degeneracy related problems. Precisely, we prove the following theorem.

Theorem 5.1. *Given an integer k and an n-vertex graph G, any p-pass streaming algorithm that can check whether $\kappa(G) \leqslant k$ or not requires $\Omega(n^2/p^3 - (n \log n)/p)$ space.*

This immediately implies the following corollary for semi-streaming algorithms.

Corollary 5.2. *A semi-streaming algorithm for any of the following problems on an n-vertex graph G requires $\tilde{\Omega}(n^{1/3})$ passes.*

- *Check whether the degeneracy $\kappa(G) \leqslant k$ for a given integer k.*
- *Exactly compute the value of $\kappa(G)$.*
- *Find a degeneracy ordering of G.*
- *Find a non-empty k-core of G for a given value of k (or report NONE if none exists).*

We prove Theorem 5.1 by reduction from BMHPC. Suppose we are given an instance of $\mathsf{BMHPC}_{m,r}$. We first describe a gadget graph G based on this instance, and prove that this graph's degeneracy $\kappa(G)$ being $\leqslant k$ or $> k$ (for some particular value k) can be used to determine whether the solution to the BMHPC instance is 0 or 1. Then, along standard lines, we show how a streaming algorithm that solves this boolean version of the degeneracy problem can be used to design a communication protocol for $\mathsf{BMHPC}_{m,r}$, implying that the space used by the streaming algorithm must be at least the communication complexity of $\mathsf{BMHPC}_{m,r}$.

5.1 Construction of the Gadget Graph

In Figure 1, we show the graph constructed for a particular instance of BMHPC. We then formally describe the construction and refer to the figure for examples.

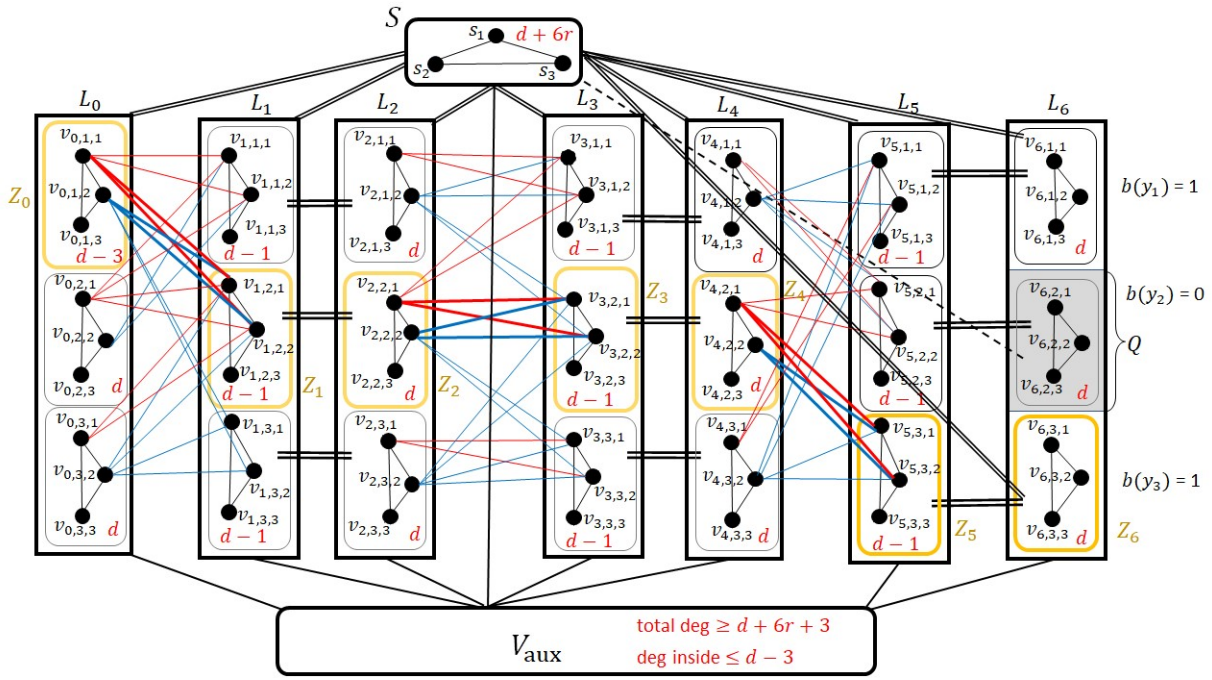


Figure 1: The gadget graph G for the following instance of $\mathsf{BMHPC}_{3,3}$. $\mathbf{A}^1 = \{A^1_{x_1} = \{y_1, y_2\}, A^1_{x_2} = \{y_1, y_2\}, A^1_{x_3} = \{y_2\}\}$. $\mathbf{B}^1 = \{B^1_{x_1} = \{y_2, y_3\}, B^1_{x_2} = \{y_1\}, B^1_{x_3} = \{y_2, y_3\}\}$. $\mathbf{C}^2 = \{C^2_{y_1} = \{x_1\}, C^2_{y_2} = \{x_1, x_2\}, C^2_{y_3} = \{x_3\}\}$. $\mathbf{D}^2 = \{D^2_{y_1} = \{x_1, x_2\}, D^2_{y_2} = \{x_2, x_3\}, D^2_{y_3} = \{x_2, x_3\}\}$. $\mathbf{A}^3 = \{A^3_{x_1} = \{y_2\}, A^3_{x_2} = \{y_2, y_3\}, A^3_{x_3} = \{y_1\}\}$. $\mathbf{B}^3 = \{B^3_{x_1} = \{y_1, y_2\}, B^3_{x_2} = \{y_3\}, B^3_{x_3} = \{y_1, y_3\}\}$ (we skip $\mathbf{C}^1, \mathbf{D}^1, \mathbf{A}^2, \mathbf{B}^2, \mathbf{C}^3, \mathbf{D}^3$ because they are redundant). Hence, $z_0 = x_1$; $z_1 = t^1_{x_1} = y_2$; $z_2 = t^2_{y_2} = x_2$; and $z_3 = t^3_{x_2} = y_3$. Therefore, the answer to this instance is $b(y_3) = 1$.
Here, a double line between two sets of nodes (e.g., the one between sets S and L_0) signifies that all possible cross edges are present between those two sets. A single line between two sets (e.g., the one between V_{aux} and L_0) denotes that some edges maybe present between them. A dashed line (e.g., the one between sets S and Q) denotes that absolutely no edge is present between them. For each triple, the number (in red) inside the box containing it denotes the degree of each node in the triple. The red and blue edges encode the inputs. The thick edges are between triples representing z_i and z_{i+1}.

The graph G has $2r + 1$ *layers* $L_0, L_1, \ldots L_{2r}$. Each layer L_ℓ has $3m$ nodes. Edges between two of these layers are always between successive layers L_ℓ and $L_{\ell+1}$ for some ℓ.

Layer nodes. For each $\ell \in [0, 2r]$ and for each $i \in [m]$, layer L_ℓ has 3 nodes (a *triple*), denoted by $v_{\ell,i,1}, v_{\ell,i,2}$, and $v_{\ell,i,3}$. For $\ell = 0$, this triple represents x_i. For $\ell \geqslant 0$, in layers $L_{4\ell+1}$ and $L_{4\ell+2}$ (resp. $L_{4\ell+3}$ and $L_{4\ell+4}$), it represents y_i (resp. x_i). For instance, in Figure 1, the triple $(v_{3,2,1}, v_{3,2,2}, v_{3,2,3})$ represents x_2 and $(v_{6,1,1}, v_{6,1,2}, v_{6,1,3})$ represents y_1. Formally, we add the following set of vertices.

$$V_L = \{v_{\ell,i,c} : \ell \in [0, 2r], i \in [m], c \in \{1, 2, 3\}\}$$

Special nodes. Outside the layers, we have a set S of three special nodes s_1, s_2, s_3.

We now define the edges on the node set $V_L \cup S$.

Edges within each triple. Each triple forms a triangle. Formally, we add the edge set E_1 to G, where

$$E_1 = \{(v_{\ell,i,1}, v_{\ell,i,2}), (v_{\ell,i,2}, v_{\ell,i,3}), (v_{\ell,i,1}, v_{\ell,i,3}) : \ell \in [0, 2r], i \in [m]\}$$

Edges between layers $L_{2\ell-1}$ and $L_{2\ell}$. As mentioned above, for each $\ell \in [r]$, either both of the ith triples in layers $L_{2\ell-1}$ and $L_{2\ell}$ represent x_i or both represent y_i. We add all 9 cross edges

between these two triples representing the same element. E.g., in Figure 1, we have all cross edges (denoted by a double line) between the triples $(v_{1,1,1}, v_{1,1,2}, v_{1,1,3})$ and $(v_{2,1,1}, v_{2,1,2}, v_{2,1,3})$, both of which represent y_1. Formally, we add the edge set E_2 to G, where

$$E_2 = \{\{v_{2\ell-1,i,c} : c \in \{1,2,3\}\} \times \{v_{2\ell,i,c} : c \in \{1,2,3\}\} : \ell \in [r], i \in [m]\}$$

Edges between layers $L_{4\ell}$ and $L_{4\ell+1}$. The edges connecting layers of the form $L_{4\ell}$ and $L_{4\ell+1}$ encode the inputs $\mathbf{A}^{2\ell+1}$ and $\mathbf{B}^{2\ell+1}$. In particular, for each $i \in [m]$, the edges leaving $v_{4\ell,i,1}$ and $v_{4\ell,i,2}$ encode the sets $A^{2\ell+1}_{x_i}$ and $B^{2\ell+1}_{x_i}$ respectively. Precisely, if the set $A^{2\ell+1}_{x_i}$ contains the element y_j, then we join $v_{4\ell,i,1}$ to both the nodes $v_{4\ell+1,j,1}$and $v_{4\ell+1,j,2}$. Similarly, if $B^{2\ell+1}_{x_i}$ contains y_j, then we join $v_{4\ell,i,2}$ to both $v_{4\ell+1,j,1}$ and $v_{4\ell+1,j,2}$ (for example, see the families ($\mathbf{A}^1$,$\mathbf{B}^1$) and ($\mathbf{A}^3$,$\mathbf{B}^3$) in Figure 1, and the corresponding edges between L_0 and L_1 and between L_4 and L_5 respectively). Formally, we add the edge sets E_A and E_B to G, where

- $E_A = \big\{\{v_{4\ell,i,1}\} \times \{v_{4\ell+1,j,1}, v_{4\ell+1,j,2}\} : y_j \in A^{2\ell+1}_{x_i}, \ell \in [0, \lfloor (r-1)/2 \rfloor], j \in [m]\big\}$
- $E_B = \big\{\{v_{4\ell,i,2}\} \times \{v_{4\ell+1,j,1}, v_{4\ell+1,j,2}\} : y_j \in B^{2\ell+1}_{x_i}, \ell \in [0, \lfloor (r-1)/2 \rfloor], j \in [m]\big\}$

Edges between layers $L_{4\ell+2}$ and $L_{4\ell+3}$. The edges between layers of the form $L_{4\ell+2}$ and $L_{4\ell+3}$ similarly encode the inputs $\mathbf{C}^{2\ell+2}$ and $\mathbf{D}^{2\ell+2}$. The construction is same as above, except with $A^{2\ell+1}_{x_i}$ and $B^{2\ell+1}_{x_i}$ replaced by $C^{2\ell+2}_{y_i}$ and $D^{2\ell+2}_{y_i}$ respectively, and y_j replaced by x_j (again, see the families $\mathbf{C}^2$ and $\mathbf{D}^2$ in the example in Figure 1 and the corresponding edges between L_2 and L_3). Formally, we add the edge sets E_C and E_D to G, where

- $E_C = \big\{\{v_{4\ell+2,i,1}\} \times \{v_{4\ell+3,j,1}, v_{4\ell+3,j,2}\} : x_j \in C^{2\ell+2}_{y_i}, \ell \in [0, \lfloor r/2 \rfloor - 1], j \in [m]\big\}$
- $E_D = \big\{\{v_{4\ell+2,i,2}\} \times \{v_{4\ell+3,j,1}, v_{4\ell+3,j,2}\} : x_j \in D^{2\ell+2}_{y_i}, \ell \in [0, \lfloor r/2 \rfloor - 1], j \in [m]\big\}$

Edges on special nodes. Recall that each element in $\mathcal{X}$ and $\mathcal{Y}$ has a bit associated with it. The special nodes s_1, s_2, s_3 are connected to each other and also to all the layer nodes except those representing bit-0-elements in the last layer L_{2r} (in Figure 1, this is the triple representing y_2 in L_6). Formally, if r is even, let $Q := \{v_{2r,i,c} : b(x_i) = 0,\ c \in \{1,2,3\}\}$, and otherwise, $Q := \{v_{2r,i,c} : b(y_i) = 0,\ c \in \{1,2,3\}\}$. Then, we add the edge set E_S to G, where

$$E_S = \{(s_1, s_2), (s_2, s_3), (s_1, s_3)\} \cup (\{s_1, s_2, s_3\} \times (V_L \setminus Q))$$

Let $V' := V_L \cup S$ and $E' := E_1 \cup E_2 \cup E_S \cup E_A \cup E_B \cup E_C \cup E_D$. The graph constructed so far is $G' = (V', E')$.

Auxiliary edges. We add some auxiliary vertices V_{aux} and edges $E_{\text{aux}} \subseteq V_{\text{aux}} \times (V' \cup V_{\text{aux}})$ to obtain the final graph

$$G = \big(V' \cup V_{\text{aux}}, E' \cup E_{\text{aux}}\big)$$

where we ensure the following degrees for $d := 6mr + 3m$ (in Figure 1, these degrees are written in the boxes containing the triples).

- $\deg(v_{0,1,1}) = \deg(v_{0,1,2}) = \deg(v_{0,1,3}) = d - 3$
- For each $i \in [2, m]$, $c \in \{1,2,3\}$, $\deg(v_{0,i,c}) = d$.
- For each $\ell \in [r], i \in [m], c \in \{1,2,3\}$, $\deg(v_{2\ell-1,i,c}) = d - 1$.
- For each $\ell \in [r], i \in [m], c \in \{1,2,3\}$, $\deg(v_{2\ell,i,c}) = d$.

- $\deg(s_1) = \deg(s_2) = \deg(s_3) = d + 6r$
- For each $u \in V_{\text{aux}}$, $\deg(u) \geqslant d + 6r + 3$.

Lemma 5.3 states that such a construction of V_{aux} and E_{aux} is possible with just the knowledge of the vertex degrees in G' (rather than the actual graph). We shall use this fact in our reduction. The proof with the explicit construction is deferred to Section D.

Lemma 5.3. *Given the degree of each node in $G' = (V', E')$, we can add vertices V_{aux} and edges $E_{aux} \subseteq V_{aux} \times (V' \cup V_{aux})$ so that the graph $G = (V' \cup V_{aux}, E' \cup E_{aux})$ has the above vertex-degrees and $\Theta(mr)$ nodes in total. Furthermore, the degree of any vertex in the induced subgraph $G[V_{aux}]$ is $\leqslant d - 3$.*

This completes the construction of the graph G.

5.2 The Analysis

In this section, we prove the following lemma that says that we can use the degeneracy value of the gadget graph constructed in the last section to determine the solution to the corresponding instance of BMHPC.

Lemma 5.4. *Given an instance of $\mathsf{BMHPC}_{m,r}$, let G denote the gadget graph constructed in Section 5.1. If $b(z_r) = 0$, then $\kappa(G) \geqslant d - 2$. Otherwise, $\kappa(G) \leqslant d - 3$.*

We set up some notations to prove the above lemma.

Triples corresponding to pointers. We are particularly interested in the triples of nodes that represent the pointers z_i in the relevant layers. In layer L_0, we are interested in the triple representing the first pointer z_0; call it Z_0. Then in *each* of the layers L_1 and L_2, we are interested in the triple representing the second pointer z_1; call these triples Z_1 and Z_2 respectively. In general, in layers $L_{2\ell-1}$ and $L_{2\ell}$, the triples that represent the pointer z_ℓ are called $Z_{2\ell-1}$ and $Z_{2\ell}$ respectively. In other words, the triple Z_ℓ corresponds to the pointer $z_{\lceil \ell/2 \rceil}$ (see Figure 1 for example. Z_0 is the triple representing $z_0 = x_1$; both Z_1 and Z_2 represent $z_1 = y_2$; both Z_3 and Z_4 represent $z_2 = x_2$; and finally both Z_5 and Z_6 represent $z_3 = y_3$.)

Let us now formalize this. Define the function $i^* : [0, 2r] \to [m]$ as follows:

For $\ell \in [0, 2r]$, let $\ell' := \lceil \ell/2 \rceil$. Then $i^*(\ell)$ is the unique index i such that

$$z_{\ell'} = \begin{cases} x_i & \text{if } \ell' \text{ is even} \\ y_i & \text{if } \ell' \text{ is odd} \end{cases}$$

Then $Z_\ell := (v_{\ell,i^*(\ell),1}, v_{\ell,i^*(\ell),2}, v_{\ell,i^*(\ell),3})$, i.e., the triple representing $z_{\lceil \ell/2 \rceil}$ in layer L_ℓ.

High-level proof overview. To prove Lemma 5.4, we perform the peeling algorithm on the graph G. We shall show that the first $6r+3$ iterations of this algorithm basically chase the hidden pointers over the successive layers: to be precise, the first three iterations remove the triple Z_0, the next three remove Z_1, and so on and so forth until iterations $6r + 1, 6r + 2$, and $6r + 3$ remove Z_{2r} (see Figures 2 to 4 for a sketch of how this proceeds). Then we have two cases as follows.

If $b(z_r) = 0$, then removal of Z_{2r} does not reduce the degrees of the nodes in S. We show that in this case, immediately after the removal of Z_{2r}, the peeling algorithm must remove a node with degree $\geqslant d - 2$ (Figure 5). Thus, by Fact 3.15, we must have $\kappa(G) \geqslant d - 2$.

Otherwise, if $b(z_r) = 1$, then Z_{2r} is indeed connected to the nodes s_1, s_2, s_3. We show that removal of Z_{2r} reduce their degrees enough so that the peeling algorithm must remove S in the next three iterations. This leads to the peeling of all the remaining nodes in the graph such that

during deletion, their degree is at most $d-3$ (see Figure 6). Then, by Fact 3.15, we must have $\kappa(G) \leqslant d-3$. This will complete the proof of the lemma.

Proof details. To formally prove Lemma 5.4, we first show that the peeling algorithm indeed peels off the Z_ℓ's one by one, and that some invariants on the vertex-degrees hold after these deletions. We start with a technical claim that we use multiple times in the proof.

Claim 5.5. *Suppose that in a graph H, the min-degree nodes are exactly a set of three nodes $U = \{u_1, u_2, u_3\}$, which are all adjacent to each other, and have degree δ each (all other nodes in H have degree $\geqslant \delta+1$). Then, the peeling algorithm on H removes the nodes in U (in some order) in the first three iterations. Further, during deletion, each u_i has degree $\leqslant \delta$.*

Proof. Since U is the exact set of min-degree nodes in H, iteration 1 of the peeling algorithm must delete one of the u_i's. WLOG, say it deletes u_1. Then, $\deg(u_2)$ and $\deg(u_3)$ drop to $\delta-1$ (since they were neighbors of u_1); at the same time, since the degree of any other node was at least $\delta+1$, it can drop to at least δ. Therefore, u_2 and u_3 remain the min-degree nodes and one of them must be removed in iteration 2. WLOG, say it's u_2. As a result, the degree of u_3 drops to $\delta-2$ (since u_2 was its neighbor) while the degree of any other node remains at least $\delta-1$. Hence, iteration 3 deletes u_3. Clearly, when each u_i was deleted, its degree was $\leqslant \delta$. □

We now list the invariants and prove that they are maintained.

Lemma 5.6. *For each $\ell \in [0, 2r]$, the following invariants hold:*

Inv. (1). Just before iteration $3\ell+1$, any $s \in S$ has $\deg(s) = d+6r-3\ell$ and any $u \in V_{aux}$ has $\deg(u) \geqslant d+6r+3-3\ell$.

Inv. (2). If ℓ is even, then just before iteration $3\ell+1$,

- *(a) each node in the triple Z_ℓ has degree $d-3$*
- *(b) for any vertex $v \in L_j$ for $j \leqslant \ell-1$, we have $\deg(v) \geqslant d-2$*
- *(c) for any vertex $v \in L_j$ for an odd $j \geqslant \ell+1$, we have $\deg(v) = d-1$*
- *(d) for any vertex $v \notin Z_\ell$ in a layer L_j for an even $j \geqslant \ell$, we have $\deg(v) = d$.*

Inv. (3). If ℓ is odd, then just before iteration $3\ell+1$,

- *(a) the triple Z_ℓ has degree $(d-3, d-3, d-1)$*
- *(b) for any vertex $v \notin Z_\ell$ in layer L_ℓ or in L_j for any $j \leqslant \ell-2$, we have $\deg(v) \geqslant d-2$*
- *(c) for any vertex $v \in L_j$ for an odd $j \geqslant \ell+2$, we have $\deg(v) = d-1$*
- *(d) for any vertex $v \in L_j$ for an even $j \geqslant \ell-1$, we have $\deg(v) = d$.*

Inv. (4). Iterations $(3\ell+1, 3\ell+2, 3\ell+3)$ delete the vertices in the triple Z_ℓ in some order. Furthermore, during deletion, each vertex has degree $\leqslant d-3$.

Proof. We prove this by induction on ℓ.

Base case. Consider the base case of $\ell = 0$. Since ℓ is even, we need to ensure Invs. (1), (2), and (4) hold. Indeed, Inv. (3) is vacuously true. Observe that by definition, $Z_0 = (v_{0,1,1}, v_{0,1,2}, v_{0,1,3})$. Before iteration 1, i.e., initially (see Figure 1), the construction ensures that each node in Z_0 has degree $d-3$. Hence, Inv. 2(a) holds. Inv. 2(b) holds vacuously. Our construction sets the degree of any vertex $v \in L_j$ for odd j to be $d-1$, and that of any vertex outside Z_0 in layer L_j for even j to be d. Thus, Inv. 2(c) and Inv. 2(d) also hold. The initial degree of any node in S is $d+6r$ and that of any node in V_{aux} is at least $d+6r+3$. This means Inv. (2) holds as well.

To see that Inv. (4) holds, note that initially the min-degree vertices in the current graph are precisely the nodes in Z_0, each with degree $d-3$. All remaining nodes have degree $\geqslant d-1$. Hence, the premise of Claim 5.5 holds with $H = G, U = Z_0$, and $\delta = d-3$. Therefore, by Claim 5.5, iterations $(1,2,3)$ delete Z_0 in some order, and during deletion, the degrees of these nodes are $\leqslant d-3$. Hence, Inv. (4) also holds, and we are done with the base case.

Inductive Step. Assume by induction hypothesis (henceforth "IH") that for some $\ell \in [0, 2r-1]$, all the invariants hold for all values $\leqslant \ell$. Then we show that they hold for $\ell+1$ as well.

First consider Inv. (1). By IH Inv. (1), any node $s \in S$ had degree $d+6r-3\ell$ before iteration $3\ell+1$. By IH Inv. (4), the next three iterations delete nodes in Z_ℓ, each of which shared an edge with each $s \in S$ (since $\ell \leqslant 2r-1$). Hence, $\deg(s)$ drops to $d+6r-3(\ell+1)$ before iteration $3(\ell+1)+1$. By the same invariant, any $u \in V_{\text{aux}}$ had degree $\geqslant d+6r+3-3\ell$ before iteration $3\ell+1$. The deletion of Z_ℓ can reduce its degree by at most 3, implying that $\deg(u) \geqslant d+6r+3-3(\ell+1)$ before iteration $3(\ell+1)+1$. Therefore, Inv. (1) holds for $\ell+1$.

For the remaining invariants, we analyze separate cases for whether $\ell+1$ is even or odd.

Case 1: $\ell+1$ is odd. We need to show that Invs. (3) and (4) hold for $\ell+1$. Inv. (2) is vacuously true.

We know that ℓ is even, and by IH, Invs. (2), and (4) hold for ℓ. By Inv. (2), we know the situation before iteration $3\ell+1$, and by Inv. (4), we know that the iterations $(3\ell+1, 3\ell+2, 3\ell+3)$ remove Z_ℓ, leading to the current graph just before iteration $3(\ell+1)+1$ (see Figure 2 for an example with $\ell = 0$ and Figure 3 for an example with $\ell = 2$ for the graph given in Figure 1).

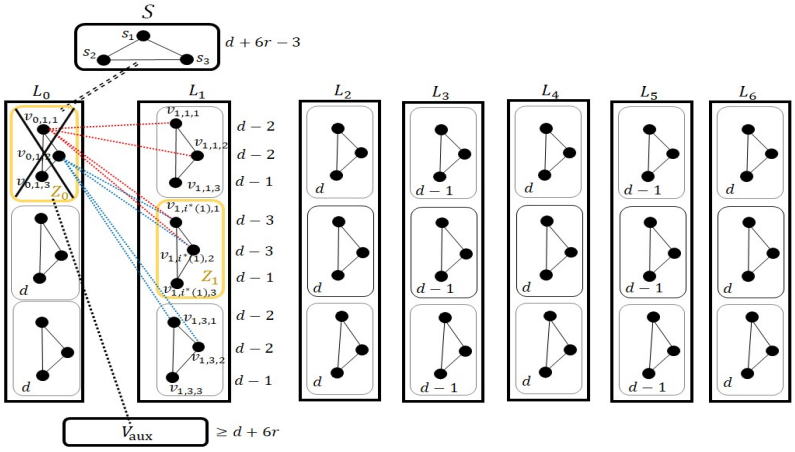


Figure 2: The status of the graph just before iteration 4. Z_0 was deleted in the last three iterations, and hence crossed out. The dotted lines denote edges that were deleted as a result. A double dotted-line between two sets of nodes signifies that originally all cross edges were present between them. Next to a node, its current degree is given. If all nodes in a set have the same degree, the value is given next to the set.

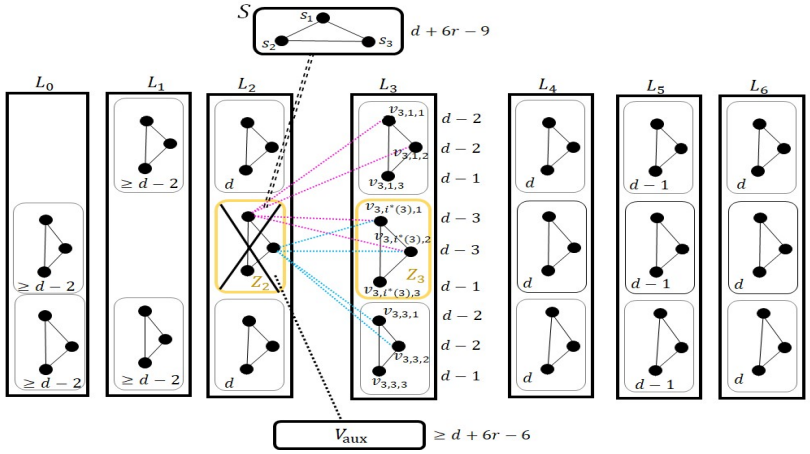


Figure 3: The status of the graph just before iteration 10. Z_2 was deleted in the last three iterations, and hence crossed out. The dotted lines denote edges that were deleted as a result. A double dotted-line between two sets of nodes signifies that originally all cross edges were present between them. Next to a node, its current degree is given. If all nodes in a set have the same degree, the value is given next to the set.

We check which nodes are affected by the removal of Z_ℓ. In the original graph, the only neighbors of Z_ℓ in layer $L_{\ell-1}$ (if exists) were the nodes in $Z_{\ell-1}$, which were removed even before Z_ℓ (by IH Inv. (4)). Thus, its only neighbors in the current graph are in layer $L_{\ell+1}$, S, or V_{aux}.

Suppose ℓ is of the form $4a+2$ (see Figure 3). Then the nodes in layer $\ell+1$ represent elements in $\mathcal{X}$. Recall that Z_ℓ corresponds to the element $z_{\ell'} \in \mathcal{Y}$ where $\ell' = \ell/2$. By construction, the edges from Z_ℓ connect to nodes that represent elements in $C^{\ell'+1}_{z_{\ell'}}$ and $D^{\ell'+1}_{z_{\ell'}}$. Consider any $x_i \in \mathcal{X}$. We have three cases: (i) x_i is in exactly one of the sets $C^{\ell'+1}_{z_{\ell'}}$ and $D^{\ell'+1}_{z_{\ell'}}$, (ii) x_i is in neither of them, and (iii) x_i is in both, i.e., $x_i = z_{\ell'+1}$. If ℓ is of the form $4a$ (see Figure 2), we make the analogous arguments replacing $\mathcal{X}$ by $\mathcal{Y}$, C and D by A and B, and x_i by y_i. We now analyze each of these cases.

By IH Inv. 2(c), before iteration $3\ell+1$, each node in the triple $(v_{\ell+1,i,1}, v_{\ell+1,i,2}, v_{\ell+1,i,3})$ had degree $d-1$. Therefore, after the removal of Z_ℓ, i.e., just before iteration $3(\ell+1)+1$, it can have the following degrees. In case (i), the triple has degrees $(d-2, d-2, d-1)$. This is because our construction ensures that exactly one of the vertices $v_{\ell,i^*(\ell),1}$ and $v_{\ell,i^*(\ell),2}$ had edges to both $v_{\ell+1,i,1}$ and $v_{\ell+1,i,2}$, and the other had none. In case (ii), it has degrees $(d-1, d-1, d-1)$. This is because none of the nodes in Z_ℓ had any edge to the triple representing x_i. In case (iii), the triple representing x_i is precisely the triple $Z_{\ell+1}$ and has degrees $(d-3, d-3, d-1)$. This follows since both nodes $v_{\ell,i^*(\ell),1}$ and $v_{\ell,i^*(\ell),2}$ had edges to both $v_{\ell+1,i,1}$ and $v_{\ell+1,i,2}$.

The above analysis immediately implies that Inv. 3(a) holds for $\ell+1$. Now, for possible values of r, consider nodes in L_r. For $r = \ell+1$, we see that any node $v \notin Z_{\ell+1}$ in layer L_r has degree $\geqslant d-2$. For $r \leqslant \ell-1 = (\ell+1)-2$, by IH Inv. 2(b), a vertex in L_r had degree $\geqslant d-2$ before iteration $3\ell+1$, which still holds after iteration $3\ell+3$: this is because, as noted above, removal of Z_ℓ does not affect any of them. Therefore, Inv. 3(b) also holds for $\ell+1$.

For an odd $j \geqslant \ell+3 = (\ell+1)+2$, by IH Inv. 2(c), any vertex in L_j had degree $d-1$ before iteration $3\ell+1$ and it is still true after iteration $3\ell+3$ since removal of Z_ℓ cannot affect them. Thus, Inv. 3(c) holds for $\ell+1$. Finally, for an even $j \geqslant \ell = (\ell+1)-1$, by IH Inv. 2(d), any vertex $v \notin Z_\ell$ had degree d before iteration $3\ell+1$; since removal of Z_ℓ doesn't affect the other vertices in these layers, their degrees still have that value and Inv. 3(d) holds for $\ell+1$. Therefore, we have shown that Inv. (3) holds for $\ell+1$.

It remains to prove Inv. (4). Since Inv. (3) holds for $\ell+1$, we know that just before iteration

$3(\ell+1)+1$, the triple $Z_{\ell+1}$ has degrees $(d-3, d-3, d-1)$, and any other node has degree at least $d-2$. This means the min-degree nodes in the current graph are in Z_ℓ; precisely, they are $v_{\ell,i^*(\ell),1}$ and $v_{\ell,i^*(\ell),2}$, each having degree $d-3$. WLOG, say iteration $3\ell+1$ deletes $v_{\ell,i^*(\ell),1}$. Then, $\deg(v_{\ell,i^*(\ell),2})$ drops to $d-4$, and $\deg(v_{\ell,i^*,3})$ drops to $d-2$. Notice that the only other nodes that can be affected by the deletion are in either layer L_ℓ or layer $L_{\ell+2}$. By Inv. 3(d) for $\ell+1$, since any node in these layers had degree d before iteration $3(\ell+1)+1$, their degree can drop to at least $d-1$ after the iteration. Also by Inv. 3, the vertices in other layers had degrees $\geqslant d-2$ before the iteration, and since they are unaffected, their degrees remain so. Hence, the current graph has the unique min-degree node $v_{\ell,i^*,2}$, which is removed by iteration $3\ell+2$. Now, $\deg(v_{\ell,i^*,3})$ drops to $d-3$. But degrees of vertices in layers L_ℓ or $L_{\ell+2}$ can now drop to at least $d-2$, while any other node maintains degree $\geqslant d-2$ as it's unaffected by the deletion. Hence, $v_{\ell,i^*,3}$ is the unique min-degree node in the current graph and is removed by iteration $3(\ell+1)+2$. Also observe that when each vertex was peeled, its degree was $\leqslant d-3$. Thus, we have proven that Inv. (4) holds for $\ell+1$.

Case 2: $\ell+1$ is even. We need to show that Invs. (2) and (4) hold for $\ell+1$. Inv. (3) holds vacuously.

We know that ℓ is odd, and by IH, Invs. (3) and (4) hold for ℓ. Analgous to the last case, by Inv. (3), we know the situation before iteration $3\ell+1$, and by Inv. (4), we know that the iterations $(3\ell+1, 3\ell+2, 3\ell+3)$ remove Z_ℓ, leading to the current graph just before iteration $3(\ell+1)+1$ (see Figure 4 for an example with $\ell=1$ for the graph in Figure 1).

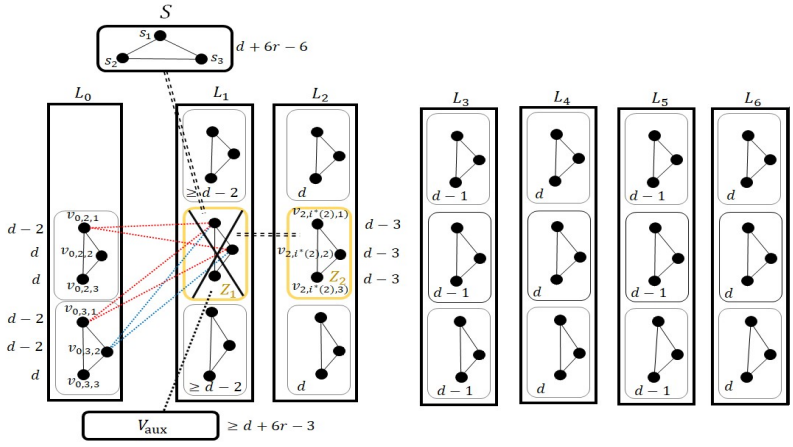


Figure 4: The status of the graph just before iteration 7. Z_1 was deleted in the last three iterations, and hence crossed out. The dotted lines denote edges that were deleted as a result. A double dotted-line between two sets of nodes signifies that originally all cross edges were present between them. Next to a node, its current degree is given. If all nodes in a set have the same degree, the value is given next to the set.

We check which nodes might be affected by the deletion of Z_ℓ in the last three iterations. By construction, Z_ℓ has edges to $Z_{\ell+1}$, to some vertices in $L_{\ell-1}$, to S, or to V_{aux}.

First consider vertices in $Z_{\ell+1}$. By IH Inv. 3(d), these nodes had degree d before iteration $3\ell+1$. Further, all of them were adjacent to each node in Z_ℓ. Hence, after Z_ℓ gets deleted, i.e., just before iteration $3(\ell+1)+1$, each of them has degree $d-3$. This shows that Inv. 2(a) holds for $\ell+1$.

Now consider the nodes in layer $L_{\ell-1}$. Suppose ℓ is of the form $4a+1$. Then the nodes in $L_{\ell-1}$ represent elements in $\mathcal{X}$ and those in L_ℓ represent elements in $\mathcal{Y}$. Recall that Z_ℓ corresponds to the element $z_{\ell'} \in \mathcal{Y}$ where $\ell' = \lceil \ell/2 \rceil$. Consider any $x_i \in \mathcal{X}$. We have three cases: (i) $z_{\ell'}$ is in exactly one of the sets $A_{x_i}^{\ell'-1}$ and $B_{x_i}^{\ell'-1}$, (ii) $z_{\ell'}$ is in neither of them, and (iii) $z_{\ell'}$ is in both, i.e., $t_{x_i}^{\ell'-1} = z_{\ell'}$.

If ℓ is of the form $4a+3$, we make the analogous arguments exchanging $\mathcal{X}$ and $\mathcal{Y}$, replacing A and B by C and D, and x_i by y_i. We analyze each case.

By IH Inv. 3(d), the triple representing x_i in layer $\ell-1$ had degree (d,d,d) before iteration $3\ell+1$. After Z_ℓ is removed, the degrees of such a triple can be the following. In case (i), it becomes either $(d-2,d,d)$ or $(d,d-2,d)$ (in Figure 4, the triple $(v_{0,2,1},v_{0,2,2},v_{0,2,3})$ is of this type). This is because exactly one of the vertices $v_{\ell-1,i,1}$ and $v_{\ell-1,i,2}$ had edges to both $v_{\ell,i^*(\ell),1}$ and $v_{\ell,i^*(\ell),2}$, and the other had none. In case (ii), it remains (d,d,d) since neither $v_{\ell-1,i,1}$ nor $v_{\ell-1,i,2}$ had any edge to Z_ℓ. In case (iii), it becomes $(d-2,d-2,d)$ (in Figure 4, the triple $(v_{0,3,1},v_{0,3,2},v_{0,3,3})$ is of this type). This follows since both nodes $v_{\ell-1,i,1}$ and $v_{\ell-1,i,2}$ had edges to both $v_{\ell,i^*(\ell),1}$ and $v_{\ell,i^*(\ell),2}$.

For possible values of j, consider the nodes in layer L_j before iteration $3(\ell+1)+1$. From the above analysis, for $j=\ell-1$, any node in L_j has degree $\geqslant d-2$. Further, by IH Inv. 3(b), for $j=\ell$ or $r\leqslant \ell-2$, any vertex $v\notin Z_\ell$ had $\deg(v)\geqslant d-2$ before iteration $3\ell+1$. Since these nodes cannot be affected by the removal of Z_ℓ, their degrees remain so. Therefore, we have shown that any vertex in layer L_j for $j\leqslant \ell=(\ell+1)-1$ has degree $\geqslant d-2$, which means Inv. 2(b) holds for $\ell+1$.

By IH Inv. 3(c), for an odd $r\geqslant \ell+2$, any vertex in L_j had degree $d-1$ before iteration $3\ell+1$. Removal of Z_ℓ doesn't affect them and hence, their degrees remain so. Thus, Inv. 2(c) holds for $\ell+1$. Finally, for an even $r\geqslant \ell+1$, IH Inv. 3(d) says that a vertex v in L_r had degree d before iteration $3\ell+1$. The deletion of Z_ℓ can only affect $Z_{\ell+1}$ among these nodes. So a vertex $v\notin Z_{\ell+1}$ in L_j still has degree d. Therefore, Inv. 2(d), and in fact, Inv. (2) holds for $\ell+1$.

It only remains to show that Inv. (4) golds for $\ell+1$. Since Inv. (2) holds for $\ell+1$, we know that just before iteration $3(\ell+1)+1$, each node in $Z_{\ell+1}$ has degree $d-3$, and any other node has degree at least $d-2$. Hence, the premise of Claim 5.5 holds with H as the current graph, $U=Z_{\ell+1}$, and $\delta=d-3$. Therefore, by Claim 5.5, the first three iterations on H, i.e., iterations $(3(\ell+1)+1,3(\ell+1)+2,3(\ell+1)+3)$ delete $Z_{\ell+1}$ in some order, and during deletion, the degree of these nodes are $\leqslant d-3$. Hence, Inv. (4) holds for $\ell+1$.

Thus, we have proven that all the invariants hold for $\ell+1$ irrespective of its parity. This completes the proof by induction. □

We are now ready to prove Lemma 5.4.

Proof of Lemma 5.4. Indeed, we have two cases.

Case 1: $b(z_r)=0$. We show that the min-degree peeling algorithm must peel a vertex of degree $\geqslant d-2$ at some iteration. Then, Fact 3.15 implies that $\kappa(G)$ must be at least $d-2$.

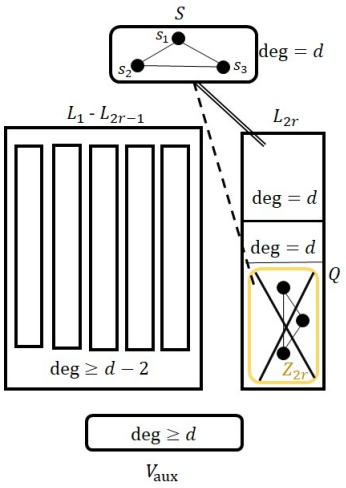


Figure 5: The graph after iteration $6r+3$ for the case $b(z_r) = 0$. A double line between two sets of nodes signifies that all possible cross edges are present between them. A dashed line denotes that no edge is present between them.

By Lemma 5.6 Inv. (4), the first $6r+3$ iterations of the peeling algorithm remove all the nodes in $Z_0, Z_1, \ldots, Z_{2r}$. Consider the status of the graph after Z_{2r} is removed (see Figure 5). By the same lemma, all nodes in layers $L_0, \ldots, L_{2r}$ had degree $\geqslant d-2$ before Z_{2r} was removed (i.e., just before iteration $6r+1$). Since its removal cannot affect any of those nodes (in these layers it only had edges to Z_{2r-1}, which was previously removed), their degrees remain so. Now consider nodes in S. By Lemma 5.6 Inv. 1, just before iteration $6r+1$, each of them had degree d. After the next 3 iterations remove Z_{2r}, their degrees remain d; this is because they were not connected to Z_{2r} as $b(z_r) = 0$. Again, by the same invariant, just before iteration $6r+1$, the nodes in V_{aux} had degree $\geqslant d+3$. Hence, after iteration $6r+3$, they must have degree $\geqslant d$. Therefore, the min-degree node in the current graph has degree $\geqslant d-2$, implying that iteration $6r+4$ of the peeling algorithm must peel a node of degree $\geqslant d-2$.

Case 2: $b(z_r) = 1$. Define the ordering σ of nodes in G as follows. It agrees with the order in which the min-degree peeling algorithm peels the nodes until iteration $6r+6$. Next, σ contains the chunk of the remaining nodes in layers $L_0, \ldots, L_{2r-1}$. They are followed by the remaining nodes in layer L_{2r}. Finally we place the auxiliary nodes V_{aux}. The vertices in each of these last three chunks can be placed in any arbitrary order within the chunk. We show that $\text{odeg}_\sigma(v) \leqslant d-3$ (see Definition 3.11) for each v.

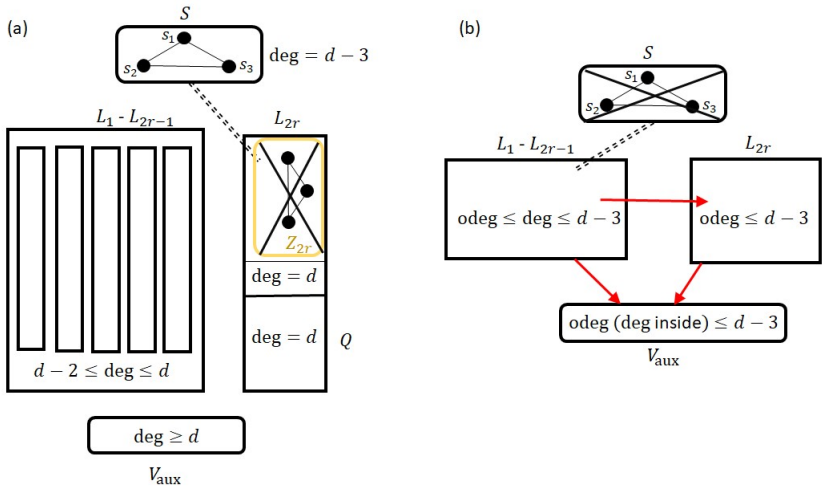


Figure 6: For the case $b(z_r) = 1$: (a) the graph after iteration $6r + 3$. Z_{2r} was deleted in the last three iterations and hence crossed out. (b) The oriented red edges show a $(d-3)$-ordering of the nodes remaining after iteration $6r + 6$. A dotted double line between two sets signifies that originally all cross edges were present between them.

As in the last case, by Inv. (4) of Lemma 5.6, the first $6r+3$ iterations of the peeling algorithm remove all nodes in $Z_0, Z_1, \ldots, Z_{2r}$. Further, each time such a node is removed, its degree in the remaining graph is at most $d-3$. Again, by Inv. (1), just before iteration $6r+1$, the nodes in S had degree d. When the next 3 iterations remove Z_{2r} (see Figure 6(a)), their degrees drop to $d-3$ this is because all nodes in S were adjacent to all nodes in Z_{2r} as $b(z_r) = 1$. By the invariants in Lemma 5.6, all other remaining nodes have degree $\geqslant d-2$. Hence, the premise of Claim 5.5 applies with H as the current graph, $U = S$, and $\delta = d-3$. By Claim 5.5, the first three iterations on H, i.e., iterations $(6r+4, 6r+5, 6r+6)$ remove the nodes in S, each having degree $\leqslant d-3$ when deleted. Since σ agrees with the peeling algorithm until this iteration, and we showed that every vertex v until now has been removed while having degree $\leqslant d-3$, it follows that $\mathrm{odeg}_\sigma(v) \leqslant d-3$ for each such v.

After S is removed (see Figure 6(b)), observe that all remaining nodes in layers $L_0, \ldots, L_{2r-1}$ have degree $\leqslant d-3$ since they had degree $\leqslant d$ before the removal of S, and they were all adjacent to all three nodes in S. Hence, regardless of their order in σ, $\mathrm{odeg}_\sigma(v) \leqslant d-3$ for any such node v. If we remove these nodes from the graph, then each node in layer L_{2r} will have degree $\leqslant d-3$ in the remaining graph: it initially had degree d, and now has lost at least its 3 neighbors in layer L_{2r-1}. Thus, $\mathrm{odeg}_\sigma(v) \leqslant d-3$ for each such node v.

Finally, the auxilary nodes V_{aux} remain, and by Lemma 5.3, the induced subgraph $G[V_{\mathrm{aux}}]$ has degree $\leqslant d-3$. Therefore, $\mathrm{odeg}_\sigma(v) \leqslant d-3$ for any $v \in V_{\mathrm{aux}}$ and we conclude that $\mathrm{odeg}_\sigma(v) \leqslant d-3$ for all $v \in V$. Hence, σ is a $(d-3)$-ordering of vertices in G. Then by Fact 3.13, we get that $\kappa(G) \leqslant d-3$. □

5.3 Completing the Reduction: Streaming to Communication Protocol

We now put things together and complete the reduction to prove Theorem 5.1. We restate the theorem here.

Theorem 5.1. *Given an integer k and an n-vertex graph G, any p-pass streaming algorithm that can check whether $\kappa(G) \leqslant k$ or not requires $\Omega(n^2/p^3 - (n \log n)/p)$ space.*

Proof. Suppose there is a p-pass S-space streaming algorithm $\mathcal{A}$ that, given an integer k, outputs

YES if $\kappa(G) \leqslant k$ and NO if not. Then, given an instance of $\mathsf{BMHPC}_{m,2p-1}$, the players P_A, P_B, P_C, and P_D run the following $(2p-1)$-phase protocol. They collectively feed the graph $G = (V, E)$ as in Lemma 5.3 to $\mathcal{A}$ as follows.

First note that the vertex set V is known to all the players. Further, the edge sets E_1, E_2, E_S are also input-independent and known to all players. In phase 1 of the protocol, P_C starts pass 1 of the algorithm $\mathcal{A}$ and inserts the edges in these sets. She then adds the edges E_C which are completely determined by her input. Next, she sends the current state of $\mathcal{A}$ along with all the vertex-degrees in the current graph to P_D. The player P_D continues the stream by adding the edges in E_D, which is determined by his input. Next, he sends the current state of $\mathcal{A}$ along with the updated vertex-degrees of the current graph (which he can calculate from the edges he added and the information sent to him by P_C) to the player P_A. This ends phase 1.

In phase 2, P_A further extends the stream by adding the edges in E_A, which is given by her input. She sends the algorithm's state along with the updated node-degrees to P_B. The player P_B adds the edge set E_B and updates the degrees. Observe that the current graph has the edge set E' as defined above. From the degree information, P_B can now determine the edges E_{aux} to be added to set the target node-degrees (by Lemma 5.3). Hence, P_B adds those edges, which completes the first pass of $\mathcal{A}$ on G. To end phase 2, P_B sends the current state of the algorithm to P_C. In phase 3, P_C starts pass 2 of the algorithm by inserting the same edges that she added in phase 1. They repeat the protocol as before for pass 2, except that they no longer communicate degree information since P_B already knows E_{aux}. Thus, the players can simulate one pass of $\mathcal{A}$ using two phases of the communication protocol, except the last pass that requires just one phase. After the final pass of $\mathcal{A}$, the player P_B looks at the output given by the algorithm for $k = d - 3 = 6m(2p-1) + 3m - 3$ and determines the answer to $\mathsf{BMHPC}_{m,2p-1}$ (using Lemma 5.4).

Thus, the players can solve $\mathsf{BMHPC}_{m,2p-1}$ using $2p-1$ phases and $O(pS + n \log n)$ bits of communication. By Theorem 4.1, it follows that $pS \geqslant \Omega(m^2 - n \log n)$. Since $n = \Theta(mp)$ by Lemma 5.3, we get that $pS \geqslant \Omega(n^2/p^2 - n \log n)$, which implies that $S \geqslant \Omega(n^2/p^3 - (n \log n)/p)$. This completes the proof of the theorem. □

A Sampling Bounds

Recall the Chernoff bound:

Proposition A.1. *Let $X_1, \cdots, X_n$ be independent random variables with values in $[0,1]$ and $X = \sum_i X_i$. Then:*

- ***Additive form:*** *For any $b \geqslant 0$, $\Pr[|X - \mathbb{E}[X]| \geqslant b] \leqslant 2 \cdot \exp\left(-\frac{b^2}{2n}\right)$,*
- ***Multiplicative form:*** *For $0 \leqslant \delta \leqslant 1$, $\Pr[|X - \mathbb{E}[X]| \geqslant (1+\delta)E[X]] \leqslant \exp\left(-\frac{\delta^2 E[X]}{3}\right)$.*

Now suppose we are allowed to repeatedly sample from an unknown Bernoulli distribution μ. We know that μ either has acceptance probability p or acceptance probability q. We wish to determine which is the case, via some method that may err with probability γ. Suppose that, for some $1 \geqslant \alpha, \beta > 0$, $p \leqslant \alpha$, whereas $q \geqslant (1+\beta)\alpha$. We may then draw

$$k = k(\alpha, \beta, \gamma) = \left\lceil \frac{48}{\alpha\beta^2} \ln\left(\frac{1}{\gamma}\right) \right\rceil$$

samples $X_1, \ldots, X_k$ from μ, and choose the threshold

$$\tau = \tau(\alpha, \beta, \gamma) = \alpha \cdot \left(1 + \frac{\beta}{4}\right) \cdot k$$

It now follows, from the multiplicative form of the Chernoff bound (Proposition A.1), that

$$\Pr[\sum_i X_i \geqslant \tau] = \Pr[\sum_i X_i \geqslant \alpha k \left(1+\frac{\beta}{4}\right)] \leqslant \exp\left(-\frac{\beta^2}{48}\alpha k\right) \leqslant \gamma.$$

Now, whenever $\beta \in [0,1]$, we have $1+\frac{\beta}{4} \leqslant (1+\beta)(1-\frac{\beta}{4})$, and so, if the acceptance probability is $q \geqslant \alpha(1+\beta)$, then

$$\begin{aligned}\Pr[\sum_i X_i \leqslant \tau] = \Pr[\sum_i X_i \leqslant \alpha k \left(1+\frac{\beta}{4}\right)] &\leqslant \Pr[\sum_i X_i \leqslant \alpha(1+\beta)k \left(1-\frac{\beta}{4}\right)] \\ &\leqslant \exp\left(-\frac{\beta^2}{48}\alpha(1+\beta)k\right) \leqslant \gamma.\end{aligned}$$

B Proof of Propositions Regarding Triangular Discrimination

Proposition 3.8. $0 \leqslant \frac{\|\mu-\nu\|_1^2}{8} \leqslant \Lambda(\mu,\nu) \leqslant \frac{\|\mu-\nu\|_1}{2} \leqslant 1$.

Proof. To see the lower bound on $\Lambda(\mu,\nu)$:

$$\frac{\|\mu-\nu\|_1}{2} = \sum_{x:\mu(x)>\nu(x)} (\mu(x)-\nu(x)) = \sum_{x:\mu(x)>\nu(x)} \frac{(\mu(x)-\nu(x))^2}{\sqrt{\mu(x)+\nu(x)}}\sqrt{\mu(x)+\nu(x)} \underset{(*)}{\leqslant} \sqrt{2\Lambda(\mu,\nu)}.$$

Here $(*)$ follows from Cauchy-Schwarz. To see the upper bound:

$$\Lambda(\mu,\nu) = \sum_{x:\mu(x)>\nu(x)} \frac{(\mu(x)-\nu(x))^2}{\mu(x)+\nu(x)} = \sum_{x:\mu(x)>\nu(x)} \frac{(\mu(x)-\nu(x))^2}{\mu(x)-\nu(x)} \cdot \frac{\mu(x)-\nu(x)}{\mu(x)+\nu(x)} \underset{(**)}{\leqslant} \frac{\|\mu-\nu\|_1}{2}.$$

Here $(**)$ follows from $\frac{\mu(x)-\nu(x)}{\mu(x)+\nu(x)} \leqslant 1$.

□

We will make use of the following:

Proposition B.1 ([Yeh20]). *If $|\eta| \leqslant \sqrt{a(b+\eta)}$ with $a,b \geqslant 0$, then $\eta \leqslant a+2b$.*

Proof. If $a=0$, then $\eta=0$ and the upper-bound follows. Otherwise, if $\eta^2 - a\eta - ab \leqslant 0$, then

$$\eta \leqslant \frac{a+\sqrt{a^2+4ab}}{2} = \frac{a}{2}(1+\sqrt{1+4b/a}) \leqslant \frac{a}{2}(1+1+4b/a).$$ □

Proposition 3.9. *For any function $f : \Omega \to \mathbb{R}_{\geqslant 0}$,*

$$\mathop{\mathbb{E}}_{\mu}[f(x)] \leqslant \Lambda(\mu,\nu) \cdot \max_{x\in\Omega} f(x) + 6 \cdot \mathop{\mathbb{E}}_{\nu}[f(x)].$$

Proof. We have:

$$\begin{aligned}|\eta| = \left|\mathop{\mathbb{E}}_{\mu}(f(x)) - \mathop{\mathbb{E}}_{\nu}(f(x))\right| &= \left|\sum_x (\mu(x)-\nu(x))f(x)\right| \\ &\leqslant \sum_x |(\mu(x)-\nu(x))|\, f(x) && \text{(By triangle inequality)} \\ &= \sum_x \frac{|\mu(x)-\nu(x)|\sqrt{f(x)}}{\sqrt{\mu(x)+\nu(x)}}\sqrt{(\mu(x)+\nu(x))f(x)} \\ &\leqslant \sqrt{\sum_x \frac{(\mu(x)-\nu(x))^2 f(x)}{\mu(x)+\nu(x)}}\sqrt{\sum_x (\mu(x)+\nu(x))f(x)} \\ & && \text{(By Cauchy-Schwarz)}\end{aligned}$$

Let us look at the first term without the square root:

$$\begin{aligned}\sum_x \frac{(\mu(x)-\nu(x))^2 f(x)}{\mu(x)+\nu(x)} &= \sum_{x:\mu(x)\geqslant\nu(x)} \frac{(\mu(x)-\nu(x))^2 f(x)}{\mu(x)+\nu(x)} + \sum_{x:\mu(x)<\nu(x)} \frac{(\mu(x)-\nu(x))^2 f(x)}{\mu(x)+\nu(x)} \\ &\leqslant \Lambda(\mu,\nu)\cdot\max_x f(x) + \sum_{x:\mu(x)<\nu(x)} \frac{(\mu(x)-\nu(x))^2 f(x)}{\mu(x)+\nu(x)} \\ &\leqslant \Lambda(\mu,\nu)\cdot\max_x f(x) + \sum_x \frac{(\nu(x))^2}{\nu(x)} f(x) = \Lambda(\mu,\nu)\cdot\max_x f(x) + \mathop{\mathbb{E}}_{\nu}(f(x)).\end{aligned}$$

which implies:

$$|\eta| = \left|\mathop{\mathbb{E}}_{\mu}(f(x)) - \mathop{\mathbb{E}}_{\nu}(f(x))\right| \leqslant \sqrt{\underbrace{(\Lambda(\mu,\nu)\cdot\max_x f(x) + \mathop{\mathbb{E}}_{\nu}(f(x)))}_{a}(\eta + \underbrace{2\mathop{\mathbb{E}}_{\nu}(f(x)))}_{b})}$$

At this point, we can use Proposition B.1 to get the following:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mu}(f(x)) - \mathop{\mathbb{E}}_{\nu}(f(x)) &\leqslant \Lambda(\mu,\nu)\cdot\max_x f(x) + 5\Pr_{\nu}(\mathcal{E}), \\ \text{i.e., } \mathop{\mathbb{E}}_{\mu}(f(x)) &\leqslant \Lambda(\mu,\nu)\cdot\max_x f(x) + 6\mathop{\mathbb{E}}_{\nu}(f(x)),\end{aligned}$$

which completes the proof. □

Proposition 3.10. *Positive triangular discrimination $\Lambda(\mu,\nu)$ is convex as a joint function of the two distributions μ and ν.*

Proof. To see this, it suffices to compute the Hessian of each term. We have

$$\Lambda(\mu,\nu) = \sum_x \begin{cases} \frac{(\mu(x)-\nu(x))^2}{\mu(x)+\nu(x)} & \text{if } \mu(x)\geqslant\nu(x) \\ 0 & \text{otherwise.}\end{cases}$$

So let us prove that each term is jointly convex in the values of $(\mu(x),\nu(x))$. We restrict our attention to the unit square. The function is zero, hence convex, inside the region where $\mu(x)\leqslant\nu(x)$. Furthermore, it is (obviously) non-negative inside the region where $\mu(x)>\nu(x)$. It is also convex in this region, since the Hessian is positive semidefinite there:

$$\nabla^2 \frac{(\mu(x)-\nu(x))^2}{\mu(x)+\nu(x)} = \begin{pmatrix} \frac{8\nu(x)^2}{(\mu(x)+\nu(x))^3} & -\frac{8\mu(x)\nu(x)}{(\mu(x)+\nu(x))^3} \\ -\frac{8\mu(x)\nu(x)}{(\mu(x)+\nu(x))^3} & \frac{8\mu(x)^2}{(\mu(x)+\nu(x))^3}\end{pmatrix}.$$

(The matrix is PSD since the off-diagonal entry, in absolute value, does not exceed the geometric mean of the diagonal entries.) It follows that each term is convex, and hence $\Lambda(\mu,\nu)$ is convex, in (μ,ν). □

C Proof of Claim 4.23

Proof. This proof is a verbatim reproduction of that in [ACK19]. For any $i\in[m]$, define $\boldsymbol{A}^{<i} := \{A_{x_1},\dots,A_{x_{i-1}}\}$, $\boldsymbol{B}^{>i} := \{B_{x_{i+1}},\dots,B_{x_m}\}$. Recall that the internal information cost of $\pi_{\mathsf{Set\text{-}Int}}$ is $\mathsf{IC}_{\mathcal{D}_{\mathsf{Set\text{-}Int}}}(\pi_{\mathsf{Set\text{-}Int}}) := \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}) + \mathbb{I}(\mathsf{B}\,;\Pi_{\mathsf{SI}} \mid \mathsf{A})$. In the following, we focus on bounding the first term. The second term can be bounded exactly the same by symmetry.

As $(\mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \boldsymbol{C}, \boldsymbol{D})$ is sampled via public randomness in $\pi_{\mathsf{Set\text{-}Int}}$, by Proposition E.7,

$$\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}) = \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \boldsymbol{C}, \boldsymbol{D}) \leqslant \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}).$$

The inequality is by Proposition E.4 as we now show $\mathsf{A} \perp (\boldsymbol{C}, \boldsymbol{D}) \mid \Pi_{\mathsf{SI}}, \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}$ (and hence conditioning on $(\boldsymbol{C}, \boldsymbol{D})$ can only decrease the mutual information). This is because $\mathsf{A} \perp (\boldsymbol{C}, \boldsymbol{D}) \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}$ by Observation 4.6 and Π_{SI} is transcript of a deterministic protocol plus $z_1, \ldots, z_j$ obtained deterministically and hence we can apply Proposition 4.7.

Define a random variable $\Theta \in \{0, 1\}$ where $\Theta = 1$ iff in Line (2) of protocol $\pi_{\mathsf{Set\text{-}Int}}$, we terminate the protocol. In other words $\Theta = 1$ iff $x_i \in Z^{<j}$. Since $\mathsf{A} \perp \Theta \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}$, further conditioning on Θ can only increase the mutual information term above by Proposition E.3, hence,

$$\begin{aligned}
\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}) &\leqslant \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta) \\
&= \frac{m-j}{m} \cdot \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) + \frac{j}{m} \cdot \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 1) \\
&\leqslant \frac{m-j}{m} \cdot \mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0), \qquad (1)
\end{aligned}$$

since conditioned on $\Theta = 1$, the protocol Π_{SI} is simple some prefix of $\mathsf{Z}^{<j}$ and is hence independent of the input (A, B) and carries no information about A (see Fact E.2-(2)). We now further bound the RHS of Eq (1). When $\Theta = 0$, $\Pi_{\mathsf{SI}} = (\mathsf{Z}^{<j}, \Pi_1, \ldots, \Pi_j) = (\mathsf{E}^{<j}, \Pi_j)$. Hence, we can write,

$$\begin{aligned}
\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) &\leqslant \mathbb{I}(\mathsf{A}\,;\mathsf{E}_j, \Pi_j \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) \\
&= \mathbb{I}(\mathsf{A}\,;\mathsf{Z}^{<j} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) \\
&\quad + \mathbb{I}(\mathsf{A}\,;\Pi^{<j}, \Pi_j \mid \mathsf{Z}^{<j}, \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) \\
&\quad \text{(by chain rule in Fact E.2-(7) and since } \mathsf{E}_j = (\Pi^{<j}, \mathsf{Z}^{<j})) \\
&\leqslant \mathbb{I}(\mathsf{A}\,;\Pi \mid \mathsf{Z}^{<j}, \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0),
\end{aligned}$$

as $\mathsf{A} \perp \mathsf{Z}^{<j} \mid \Theta = 0$ (and other variables) and hence the first term is zero, and in the second term Π contains $\Pi^{<j}, \Pi_j$ (plus potentially other terms) and so having Π in instead can only increase the

information. By further expanding the conditional information term above,

$$\begin{aligned}
&\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}, \mathsf{I}, \boldsymbol{A}^{<\mathsf{I}}, \boldsymbol{B}^{>\mathsf{I}}, \Theta = 0) \\
&\qquad \leqslant \mathop{\mathbb{E}}_{(Z^{<j},i)|\Theta=0} \left[\mathbb{I}(\mathsf{A}\,;\Pi \mid \mathsf{B}, \boldsymbol{A}^{<i}, \boldsymbol{B}^{>i}, \mathsf{I} = i, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0)\right] \\
&\qquad = \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\sum_{\substack{i=1 \\ i \notin Z^{<j}}}^{m} \frac{1}{m-j} \mathbb{I}(\mathsf{A}_{x_i}\,;\Pi \mid \mathsf{B}_{x_i}, \boldsymbol{A}^{<i}, \boldsymbol{B}^{>i}, \mathsf{I} = i, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0) \right] \\
&\qquad\qquad \text{(conditioned on } \Theta = 0,\ i \text{ is chosen uniformly at random from } Z^{<j}\text{; also } (\mathsf{A}, \mathsf{B}) = (\mathsf{A}_{x_i}, \mathsf{B}_{x_i})) \\
&\qquad = \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\sum_{i \notin Z^{<j}} \frac{1}{m-j} \cdot \mathbb{I}(\mathsf{A}_{x_i}\,;\Pi \mid \mathsf{B}_{x_i}, \boldsymbol{A}^{<i}, \boldsymbol{B}^{>i}, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0) \right] \\
&\qquad\qquad \text{(we dropped the conditioning on } \mathsf{I} = i \text{ as all remaining variables are independent of this event)} \\
&\qquad = \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\sum_{i \notin Z^{<j}} \frac{1}{m-j} \cdot \mathbb{I}(\mathsf{A}_{x_i}\,;\Pi \mid \boldsymbol{A}^{<i}, \boldsymbol{B}, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0) \right] \\
&\qquad\qquad \text{(as } \mathsf{A}_{x_i} \perp \boldsymbol{B}^{<i} \mid \mathsf{B}_{x_i}, \boldsymbol{A}^{<i} \text{ by Observation 4.6 and hence we can apply Proposition E.3)} \\
&\qquad \leqslant \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\sum_{i=1}^{m} \frac{1}{m-j} \cdot \mathbb{I}(\mathsf{A}_{x_i}\,;\Pi \mid \boldsymbol{A}^{<i}, \boldsymbol{B}, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0) \right] \\
&\qquad\qquad \text{(mutual information is non-negative by Fact E.2-(2) and so we can add the terms in } Z^{<j} \text{ as well)} \\
&\qquad = \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\sum_{i=1}^{m} \frac{1}{m-j} \cdot \mathbb{I}(\mathsf{A}_{x_i}\,;\Pi \mid \boldsymbol{A}^{<i}, \boldsymbol{B}, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0) \right] \\
&\qquad = \frac{1}{m-j} \cdot \mathop{\mathbb{E}}_{Z^{<j}|\Theta=0} \left[\mathbb{I}(\boldsymbol{A}\,;\Pi \mid \boldsymbol{B}, \mathsf{Z}^{<j} = Z^{<j}, \Theta = 0)\right] \quad \text{(by chain rule in Fact E.2-(7))} \\
&\qquad = \frac{1}{m-j} \cdot \mathbb{I}(\boldsymbol{A}\,;\Pi \mid \boldsymbol{B}, \mathsf{Z}^{<j}, \Theta = 0) \\
&\qquad \leqslant \frac{1}{m-j} \cdot \left(\mathbb{I}(\boldsymbol{A}\,;\Pi \mid \boldsymbol{B}, \Theta = 0) + \mathbb{H}(\mathsf{Z}^{<j})\right) \\
&\qquad = \frac{1}{m-j} \cdot \left(\mathbb{I}(\boldsymbol{A}\,;\Pi \mid \boldsymbol{B}) + \mathbb{H}(\mathsf{Z}^{<j})\right) \\
&\qquad\qquad \text{(transcript of the protocol } \pi_{\mathsf{BHPC}} \text{ (namely } \Pi\text{) on input } (\boldsymbol{A}, \boldsymbol{B}) \text{ is independent of } \Theta) \\
&\qquad \leqslant \frac{1}{m-j} \cdot \left(\mathbb{H}(\Pi) + \mathbb{H}(\mathsf{Z}^{<j})\right) \leqslant \frac{\mathsf{CC}(\pi_{\mathsf{BHPC}})}{m-j} + \frac{j \cdot \log m}{m-j}. \\
&\qquad\qquad \text{(by sub-additivity of entropy (Fact E.2-(5)) and Fact E.2-(1))}
\end{aligned}$$

By plugging in this bound in Eq (1), we have that,

$$\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}) \leqslant \frac{m-j}{m} \cdot \left(\frac{\mathsf{CC}(\pi_{\mathsf{BHPC}})}{m-j} + \frac{j \cdot \log m}{m-j}\right) = \frac{\mathsf{CC}(\pi_{\mathsf{BHPC}})}{m} + \frac{j \cdot \log m}{m}.$$

By symmetry, we can also prove the same bound on $\mathbb{I}(\mathsf{B}\,;\Pi_{\mathsf{SI}} \mid \mathsf{A})$. As such, we have,

$$\mathbb{I}(\mathsf{A}\,;\Pi_{\mathsf{SI}} \mid \mathsf{B}) + \mathbb{I}(\mathsf{B}\,;\Pi_{\mathsf{SI}} \mid \mathsf{A}) \leqslant 2 \cdot \left(\frac{\mathsf{CC}(\pi_{\mathsf{BHPC}})}{m} + \frac{j \cdot \log m}{m}\right).$$

We shall note that strictly speaking the factor 2 above is not needed (similar to the proof of Proposition E.8) but as this factor is anyway suppressed through O-notation later in the proof, the above bound suffices for our purpose.

□

D Proof of Lemma 5.3

Consider the graph $G' = (V', E')$. Set $d = 6mr + 3m$. We add d auxiliary nodes in V_{aux}. Since the target degree of each node outside S is at most d, and the nodes in S need at most $6r + 3m \leqslant d$ edges to reach target degree, the d auxiliary nodes suffice to increase their degree. Thus, the total number of vertices in G is $|V'| + d = \Theta(mr)$.

We now describe the construction. For each node $v \in V'$, define its deficiency $\texttt{def}(v)$ as the difference between its target degree mentioned above and its current degree in G'. Define the total deficiency $D := \sum_{v \in V'} \texttt{def}(v)$. Fix arbitrary orderings $v_1, v_2, \ldots v_{|V'|}$ of the nodes in V' and $u_1, \ldots, u_d$ of the nodes in V_{aux}. From each v_i, we join edges to $\texttt{def}(v_i)$ many u_j's following the ordering. The first neighbor of v_{i+1} is taken to be the node that is next to the last neighbor of v_i in the ordering of the u_j's. We wrap around to u_1 after u_d. Hence, the degree of two vertices in V_{aux} can differ by at most 1. We check the min-degree of a node in V_{aux}. If it's $\geqslant d + 6r + 3$, then we are done. Otherwise, suppose it's $d + 6r + 3 - x$ for some $x > 0$. Then we add x disjoint matchings of size $d/2$ in V_{aux} to increase the degree of each node by x, i.e., to ensure that each node in V_{aux} has degree $\geqslant d + 6r + 3$.

It remains to prove the "furthermore" part. For this, we check the maximum and minimum deficiency of each node.

Observe that in G', each node in layer 0 has degree at least 5 (it's adjacent to the other 2 nodes in its triple and to the 3 nodes in S) and at most $2m + 5$ (for each $i \in [m]$, it is adjacent to at most 2 nodes that represent i in layer 1). Hence, the deficiency of any such node is in $[d - 2m - 5, d - 5]$.

For any node in a layer $\ell \in [2r - 1]$, it is adjacent to at least 8 nodes (the other 2 nodes in its triple, the 3 nodes in S, and the 3 nodes representing the same element in its "paired layer", i.e., layer $\ell + 1$ if ℓ is odd, or layer $\ell - 1$ if it's even) and at most $2m + 8$ nodes (for each $i \in [m]$, it is adjacent to at most 2 nodes that represent i in layer $\ell - 1$ if ℓ is odd, or layer $\ell + 1$ if it's even). Thus, the deficiency of any such node is in $[d - 2m - 8, d - 8]$.

Any node in layer $2r$ representing x_i with $b(x_i) = 0$ has degree exactly 5 (the 2 nodes in its triple and the 3 nodes representing the same element in layer $2r - 1$). Finally, any other nodes in layer $2r$ has degree exactly 8 (same as other nodes in layer $2r$ with the addition of the 3 nodes in S). Hence, the deficiency of any such node is in $[d - 8, d - 5]$.

Therefore, the total deficiency D is at least $(d - 2m - 8) \cdot |V_L| = (d - 2m - 8) \cdot d \geqslant 9d^2/10$ for large enough m. These edges are received almost equally by the d nodes in $|V_{\text{aux}}|$. Thus, the min-degree in $|V_{\text{aux}}|$ after the addition of these edges is at least $9d/10$. Thus the value of x is at most $d + 6r + 3 - 9d/10 \leqslant d - 3$ (again for large enough m). The x matchings can induce degree at most $x \leqslant d - 3$ in the subgraph $G[V_{\text{aux}}]$, and hence we are done with the furthermore part.

E Background on Information Theory

We now briefly introduce some definitions and facts from information theory that are needed in this thesis. We refer the interested reader to the text by Cover and Thomas [CT06] for an excellent introduction to this field, and the proofs of the statements used in this Appendix.

For a random variable A, we use $\text{SUPP}(\mathsf{A})$ to denote the support of A and $\text{dist}(\mathsf{A})$ to denote its distribution. When it is clear from the context, we may abuse the notation and use A directly instead of $\text{dist}(\mathsf{A})$, for example, write $A \sim \mathsf{A}$ to mean $A \sim \text{dist}(\mathsf{A})$, i.e., A is sampled from the distribution of random variable A.

We denote the *Shannon Entropy* of a random variable A by $\mathbb{H}(\mathsf{A})$, which is defined as:

$$\mathbb{H}(\mathsf{A}) := \sum_{A \in \text{SUPP}(\mathsf{A})} \Pr(\mathsf{A} = A) \cdot \log(1/\Pr(\mathsf{A} = A)) \tag{2}$$

The *conditional entropy* of A conditioned on B is denoted by $\mathbb{H}(\mathsf{A} \mid \mathsf{B})$ and defined as:

$$\mathbb{H}(\mathsf{A} \mid \mathsf{B}) := \mathop{\mathbb{E}}_{B \sim \mathsf{B}} [\mathbb{H}(\mathsf{A} \mid \mathsf{B} = B)], \tag{3}$$

where $\mathbb{H}(\mathsf{A} \mid \mathsf{B} = B)$ is defined in a standard way by using the distribution of A conditioned on the event $\mathsf{B} = B$ in Eq (2).
Recall that the binary entropy $H(p)$ is the entropy of a two-valued random variable that takes one value with probability p. We need the following upper bound on the binary entropy.

Fact E.1. *The binary entropy* $H_2(p) \leqslant 2\sqrt{p \cdot (1-p)}$.

The *mutual information* of two random variables A and B is denoted by $\mathbb{I}(\mathsf{A}\,;\mathsf{B})$ and is defined as:

$$\mathbb{I}(\mathsf{A}\,;\mathsf{B}) := \mathbb{H}(A) - \mathbb{H}(A \mid B) = \mathbb{H}(B) - \mathbb{H}(B \mid A). \tag{4}$$

The *conditional mutual information* $\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C})$ is $\mathbb{H}(\mathsf{A} \mid \mathsf{C}) - \mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C})$ and hence by linearity of expectation:

$$\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) = \mathop{\mathbb{E}}_{C \sim \mathsf{C}} [\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C} = C)]. \tag{5}$$

E.1 Useful Properties of Entropy and Mutual Information

We shall use the following basic properties of entropy and mutual information throughout.

Fact E.2. *Let* A, B, C, *and* D *be four (possibly correlated) random variables.*

1. $0 \leqslant \mathbb{H}(\mathsf{A}) \leqslant \log|\text{SUPP}(\mathsf{A})|$. *The right equality holds iff* $\text{dist}(\mathsf{A})$ *is uniform.*
2. $\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) \geqslant 0$. *The equality holds iff* A *and* B *are* independent *conditioned on* C.
3. $\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) \leqslant \mathbb{H}(\mathsf{A})$.
4. Conditioning on a random variable reduces entropy*:* $\mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}) \leqslant \mathbb{H}(\mathsf{A} \mid \mathsf{B})$. *The equality holds iff* $\mathsf{A} \perp \mathsf{C} \mid \mathsf{B}$.
5. Subadditivity of entropy*:* $\mathbb{H}(\mathsf{A}, \mathsf{B} \mid \mathsf{C}) \leqslant \mathbb{H}(\mathsf{A} \mid C) + \mathbb{H}(\mathsf{B} \mid \mathsf{C})$.
6. Chain rule for entropy*:* $\mathbb{H}(\mathsf{A}, \mathsf{B} \mid \mathsf{C}) = \mathbb{H}(\mathsf{A} \mid \mathsf{C}) + \mathbb{H}(\mathsf{B} \mid \mathsf{C}, \mathsf{A})$.
7. Chain rule for mutual information*:* $\mathbb{I}(\mathsf{A}, \mathsf{B}\,;\mathsf{C} \mid \mathsf{D}) = \mathbb{I}(\mathsf{A}\,;\mathsf{C} \mid \mathsf{D}) + \mathbb{I}(\mathsf{B}\,;\mathsf{C} \mid \mathsf{A}, \mathsf{D})$.
8. Data processing inequality*: for a function* $f(\mathsf{A})$ *of* A, $\mathbb{I}(f(\mathsf{A})\,;\mathsf{B} \mid \mathsf{C}) \leqslant \mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C})$.

We also use the following two standard propositions, regarding the effect of conditioning on mutual information.

Proposition E.3. *For random variables* $\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D}$, *if* $\mathsf{A} \perp \mathsf{D} \mid \mathsf{C}$, *then,*

$$\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) \leqslant \mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}, \mathsf{D}).$$

Proof. Since A and D are independent conditioned on C, by Fact E.2-(4), $\mathbb{H}(\mathsf{A} \mid \mathsf{C}) = \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{D})$ and $\mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{B}) \geqslant \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{B}, \mathsf{D})$. We have,

$$\begin{aligned}\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) &= \mathbb{H}(\mathsf{A} \mid \mathsf{C}) - \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{B}) = \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{D}) - \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{B}) \\ &\leqslant \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{D}) - \mathbb{H}(\mathsf{A} \mid \mathsf{C}, \mathsf{B}, \mathsf{D}) = \mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}, \mathsf{D}). \quad \square\end{aligned}$$

□

Proposition E.4. *For random variables* $\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D}$*, if* $\mathsf{A} \perp \mathsf{D} \mid \mathsf{B}, \mathsf{C}$*, then,*

$$\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) \geqslant \mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}, \mathsf{D}).$$

Proof. Since $\mathsf{A} \perp \mathsf{D} \mid \mathsf{B}, \mathsf{C}$, by Fact E.2-(4), $\mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}) = \mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}, \mathsf{D})$. Moreover, since conditioning can only reduce the entropy (again by Fact E.2-(4)),

$$\begin{aligned}\mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}) &= \mathbb{H}(\mathsf{A} \mid \mathsf{C}) - \mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}) \geqslant \mathbb{H}(\mathsf{A} \mid \mathsf{D}, \mathsf{C}) - \mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}) \\ &= \mathbb{H}(\mathsf{A} \mid \mathsf{D}, \mathsf{C}) - \mathbb{H}(\mathsf{A} \mid \mathsf{B}, \mathsf{C}, \mathsf{D}) = \mathbb{I}(\mathsf{A}\,;\mathsf{B} \mid \mathsf{C}, \mathsf{D}). \quad \square\end{aligned}$$

□

E.2 Background on Communication and Information Complexity

Communication complexity. We briefly review the standard definitions of the two-party communication model of Yao [Yao79]. See the text by Kushilevitz and Nisan [KN97] for an extensive overview of communication complexity. In Section 4, we also use a standard generalization of this model to allow for more than two players, but we defer the necessary definitions to that section.

Let $P : \mathcal{X} \times \mathcal{Y} \to \mathcal{Z}$ be a relation. Alice receives an input $X \in \mathcal{X}$ and Bob receives $Y \in \mathcal{Y}$, where (X, Y) are chosen from a joint distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$. We allow players to have access to both public and private randomness. They communicate with each other by exchanging messages such that each message depends only on the private input and random bits of the player sending the message, and the already communicated messages plus the public randomness. At the end, one of the players need to output an answer Z such that $Z \in P(X, Y)$.

We use π to denote a protocol used by the players. We always assume that the protocol π can be randomized (using both public and private randomness), *even against a prior distribution* $\mathcal{D}$ *of inputs*. For any $0 < \delta < 1$, we say π is a δ-error protocol for P over a distribution $\mathcal{D}$, if the probability that for an input (X, Y), π outputs some Z where $Z \notin P(X, Y)$ is at most δ (the probability is taken over the randomness of *both* the distribution and the protocol).

Definition E.5 (Communication cost)**.** The *communication cost* of a protocol π on an input distribution $\mathcal{D}$, denoted by $\mathsf{CC}_{\mathcal{D}}(\pi)$, is the *worst-case* bit-length of the transcript communicated between Alice and Bob in the protocol π, when the inputs are chosen from $\mathcal{D}$.

Communication complexity of a problem P is defined as the minimum communication cost of a protocol π that solves P on every distribution $\mathcal{D}$ with probability at least $2/3$.

Information complexity. There are several possible definitions of information cost of a communication prtocol that have been considered depending on the application (see, e.g., [CSWY01, BJKS02, BBCR10, BR11, BEO+13]). We use the notion of *internal information cost* [BBCR10] that measures the average amount of information each player learns about the input of the other player by observing the transcript of the protocol.

Definition E.6 (Information cost)**.** Consider an input distribution $\mathcal{D}$ and a protocol π. Let $(\mathsf{X},\mathsf{Y}) \sim \mathcal{D}$ denote the random variables for the input of Alice and Bob and Π be the the random variable for the transcript of the protocol *concatenated* with the public randomness R used by π. The *(internal) information cost* of π with respect to $\mathcal{D}$ is $\mathsf{IC}_{\mathcal{D}}(\pi) := \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{X} \mid \mathsf{Y}) + \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{Y} \mid \mathsf{X})$.

One can also define information complexity of a problem P similar to communication complexity with respect to the information cost. However, we avoid presenting this definition formally due to some subtle technical issues that need to be addressed which lead to multiple different but similar-in-spirit definitions. As such, we state our results directly in terms of information cost.

Note that any public coin protocol is a distribution over private coins protocols, run by first using public randomness to sample a random string $\mathsf{R} = R$ and then running the corresponding private coin protocol π^R. We also use Π^R to denote the transcript of the protocol π^R. We have the following standard proposition.

Proposition E.7. *For any distribution $\mathcal{D}$ and any protocol π with public randomness $\boldsymbol{R}$,*

$$\mathsf{IC}_{\mathcal{D}}(\pi) = \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{X} \mid \mathsf{Y},\mathsf{R}) + \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{Y} \mid \mathsf{X},\mathsf{R}) = \mathop{\mathbb{E}}_{R\sim\mathsf{R}}\left[\mathsf{IC}_{\mathcal{D}}(\pi^R)\right].$$

Proof. By definition of internal information cost,

$$\begin{aligned}
\mathsf{IC}_{\mathcal{D}}(\pi) &= \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{X} \mid \mathsf{Y}) + \mathbb{I}_{\mathcal{D}}(\Pi\,;\mathsf{Y} \mid \mathsf{X}) = \mathbb{I}(\Pi,\mathsf{R}\,;\mathsf{X} \mid \mathsf{Y}) + \mathbb{I}(\Pi,\mathsf{R}\,;\mathsf{Y} \mid \mathsf{X}) \\
&\qquad\qquad \text{(Π denotes the transcript and the public randomness)} \\
&= \mathbb{I}(\mathsf{R}\,;\mathsf{X} \mid \mathsf{Y}) + \mathbb{I}(\Pi\,;\mathsf{X} \mid \mathsf{Y},\mathsf{R}) + \mathbb{I}(\mathsf{R}\,;\mathsf{Y} \mid \mathsf{X}) + \mathbb{I}(\Pi\,;\mathsf{Y} \mid \mathsf{X},\mathsf{R}) \\
&\qquad\qquad \text{(chain rule of mutual information, Fact E.2-(7))} \\
&= \mathbb{I}(\Pi\,;\mathsf{X} \mid \mathsf{Y},\mathsf{R}) + \mathbb{I}(\Pi\,;\mathsf{Y} \mid \mathsf{X},\mathsf{R}) \\
&\qquad\qquad \text{($\mathbb{I}(\mathsf{R}\,;\mathsf{X} \mid \mathsf{Y}) = \mathbb{I}(\mathsf{R}\,;\mathsf{Y} \mid \mathsf{X}) = 0$ since $\mathsf{R} \perp \mathsf{X},\mathsf{Y}$ and Fact E.2-(2))} \\
&= \mathop{\mathbb{E}}_{R\sim\mathsf{R}}\left[\mathbb{I}(\Pi\,;\mathsf{X} \mid \mathsf{Y},\mathsf{R}=R) + \mathbb{I}(\Pi\,;\mathsf{Y} \mid \mathsf{X},\mathsf{R}=R)\right] = \mathop{\mathbb{E}}_{R\sim\mathsf{R}}\left[\mathsf{IC}_{\mathcal{D}}(\pi^R)\right],
\end{aligned}$$

concluding the proof. □

The following well-known proposition relates communication cost and information cost.

Proposition E.8 (cf. [BR11])**.** *For any distribution $\mathcal{D}$ and any protocol π: $\mathsf{IC}_{\mathcal{D}}(\pi) \leqslant \mathsf{CC}_{\mathcal{D}}(\pi)$.*

Proof. Let us assume first that π only uses private randomness and thus Π only contain the transcript. For any $b \in [\mathsf{CC}_{\mathcal{D}}(\pi)]$, we define Π_b to be the b-th bit of the transcript. We have,

$$\begin{aligned}
\mathsf{IC}_{\mathcal{D}}(\pi) &= \mathbb{I}(\Pi\,;\mathsf{X} \mid \mathsf{Y}) + \mathbb{I}(\Pi\,;\mathsf{Y} \mid \mathsf{X}) \\
&= \sum_{b=1}^{\mathsf{CC}_{\mathcal{D}}(\pi)} \mathbb{I}(\Pi_b\,;\mathsf{X} \mid \Pi^{<b},\mathsf{Y}) + \mathbb{I}(\Pi_b\,;\mathsf{Y} \mid \Pi^{<b},\mathsf{X}) \\
&\qquad\qquad \text{(by chain rule of mutual information in Fact E.2-(7))} \\
&= \sum_{b=1}^{\mathsf{CC}_{\mathcal{D}}(\pi)} \mathop{\mathbb{E}}_{\Pi^{<b}}\left[\mathbb{I}(\Pi_b\,;\mathsf{X} \mid \Pi^{<b}=\Pi^{<b},\mathsf{Y}) + \mathbb{I}(\Pi_b\,;\mathsf{Y} \mid \Pi^{<b}=\Pi^{<b},\mathsf{X})\right].
\end{aligned}$$

Consider each term in the RHS above. By conditioning on $\Pi^{<b}$, the player that transmit Π_b would become fix. If this player is Alice, then $\mathbb{I}(\Pi_b\,;\mathsf{Y} \mid \Pi^{<b}=\Pi^{<b},\mathsf{X}) = 0$, because Π_b is only a function of $(\Pi^{<b},\mathsf{X})$ in this case; similarly, if this player is Bob, then $\mathbb{I}(\Pi_b\,;\mathsf{X} \mid \Pi^{<b}=\Pi^{<b},\mathsf{Y}) = 0$. Moreover, $\mathbb{I}(\Pi_b\,;\mathsf{X} \mid \Pi^{<b}=\Pi^{<b},\mathsf{Y}) \leqslant \mathbb{H}(\Pi_b) \leqslant 1$ and similarly $\mathbb{I}(\Pi_b\,;\mathsf{Y} \mid \Pi^{<b}=\Pi^{<b},\mathsf{X}) \leqslant 1$. As such, the above term can be upper bounded by $\mathsf{CC}_{\mathcal{D}}(\pi)$. To finalize the proof, note that by Proposition E.7, for any public-coin protocol π, $\mathsf{IC}_{\mathcal{D}}(\pi) = \mathbb{E}_{R\sim\mathsf{R}}\left[\mathsf{IC}_{\mathcal{D}}(\pi^R)\right] \leqslant \mathbb{E}_{R\sim\mathsf{R}}\left[\mathsf{CC}_{\mathcal{D}}(\pi^R)\right] \leqslant \mathsf{CC}_{\mathcal{D}}(\pi)$, where the first inequality is by the first part of the argument. □

Proposition E.8 provides a convinent way of proving communication complexity lower bounds by lower bounding information cost of any protocol.

Rectangle Property of Communication Protocols

We conclude this section by mentioning some basic properties of communication protocols. For any protocol π and inputs $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, we define $\Pi_{x,y}$ as the transcript of the protocol conditioned on the input x to Alice and input y to Bob. Note that for randomized protocols, $\Pi_{x,y}$ is a random variable which we denote by $\mathsf{\Pi}_{x,y}$.

The following is referred to as the rectangle property of deterministic protocols.

Fact E.9 (Rectangle property)**.** *For any deterministic protocol π and inputs $x, x' \in \mathcal{X}$ to Alice and $y, y' \in \mathcal{Y}$ to Bob, if $\Pi_{x,y} = \Pi_{x',y'}$, then $\Pi_{x,y'} = \Pi_{x',y}$.*

Fact E.9 implies that the set of inputs consistent with any transcript $\Pi_{x,y}$ of a deterministic protocol forms a combinatorial rectangle. One can also extend the rectangle property of deterministic protocols to randomized protocols using the following fact.

Fact E.10 (Cut-and-paste property; cf. [BJKS02])**.** *For any randomized protocol π and inputs $x, x' \in \mathcal{X}$ to Alice and $y, y' \in \mathcal{Y}$ to Bob, $\mathrm{h}(\mathsf{\Pi}_{x,y}, \mathsf{\Pi}_{x',y'}) = \mathrm{h}(\mathsf{\Pi}_{x,y'}, \mathsf{\Pi}_{x',y})$.*

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